RAAM

Regime-Aware Asset Management

QWIM Market Regime Detection and Dynamic Portfolio Allocation

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Medium-Long Term Wealth Management Application

Group 1:

Mina Fahim | Nabil Salame | Willie Gadson

Niti Bhavesh Shah | Ojaus Mane | Darshan Nanjegowda

GitHub Repository:

https://github.com/mina-fahim/FE800-Market-Regime-Detection

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ABSTRACT

Financial markets are non-stationary, shifting between different states due to economic and non-economic factors. A static portfolio is problematic because in a bull market, the investor may miss the opportunity to achieve better risk-adjusted returns by keeping the same portfolio allocation across different regimes. On the other hand, investors can suffer large losses during bear markets such as the 2008 financial crisis or COVID-19.

To detect which regime we are in, we built 4 different models: Hidden Markov Model, Semi-Hidden Markov Model, Continuous Jump Model, and Machine Learning Model. Once the regime is detected, we use a constrained mean-variance optimization to allocate according to the risk profile of each regime. We benchmark the results against a static portfolio and a mean-variance optimized portfolio. The static portfolio allocation is 55% equity, 40% bonds, and 5% alternatives (gold).

After testing the models on data from 2001 to 2025 using a walk-forward method, training from 2001 to 2018, then testing from 2019 through 2025, the machine learning and jump models outperformed the static benchmark annually with higher Sharpe ratios and better Sortino ratios. The machine learning model had better maximum drawdown control, while the jump model had lower turnover, switching only 8 times compared to 34 times for the machine learning model.

The choice of model depends on the investor's risk tolerance. Investors with higher tolerance will prefer the jump model since it switches less, leading to lower turnover and lower transaction costs. Investors with lower tolerance will prefer the machine learning model as it responds faster when a new regime is detected, reducing exposure to market swings.

One key limitation is the 2022 period, which broke historical patterns due to unusually high inflation and simultaneous asset price declines. As more data becomes available, the models may be able to identify such a regime and allocate accordingly.

1. INTRODUCTION

Financial markets do not follow a single, stable process. Although there is ample evidence of volatility clustering, shifting correlations, and sudden shifts in return distributions across asset classes, static allocation, a fixed portfolio retained regardless of the current economic climate remains the predominant approach in wealth management. The 2008 financial crisis and the 2020 COVID-19 shock showed how rapidly static portfolios can experience substantial, protracted drawdowns, demonstrating that this gap between market reality and portfolio creation is more than just a theoretical issue.

The academic case for regime-based investing is already established. Nystrup et al. (2017) showed that switching between risk-on and risk-off positioning across detected market states improves performance, and Aydınhan et al. (2024) demonstrated the advantage of probabilistic regime assignment over hard clustering. Yet most existing frameworks are limited in scope: binary risk-on/off switches, single-model detection, or narrow asset universes, leaving a gap between the theoretical regime-aware allocation and a practical, multi-asset implementation.

This paper addresses exactly that gap. We develop RAAM (Regime-Aware Asset Management), a framework that combines a rich 49-feature macroeconomic and market signal matrix with four regime detection models: Hidden Markov, Hidden Semi-Markov, Continuous Jump, and a GMM-XGBoost hybrid. Each of these models feeds into a constrained mean-variance optimizer calibrated to the risk profile of each detected regime. The framework is designed for medium-to-long-term retail wealth management and is validated through a walk-forward backtest to prove it performs well out-of-sample.

Our primary contribution is not one single model, but the systematic comparison of structurally different regime identification approaches within a unified portfolio construction and evaluation pipeline, allowing investors to select a model aligned with their tolerance for turnover, responsiveness, and drawdown risk.

2. LITERATURE REVIEW

Nystrup, Madsen, and Lindstrom (2017) showed that allocating differently across different market regimes works effectively. However, it was buy and hold versus risk on by buying the index and risk off by staying in cash. We extended their model by adding a transition regime and allocating to a multi-asset portfolio.

Baltussen et al. (2023) studied the effect of inflation on several asset returns across different regimes, where assets' real returns alter across different inflation environments. This motivated us to include inflation indicators such as CPI and CPI change as part of our feature matrix, as well as commodities like oil to capture inflationary pressure.

Aydınhan et al. (2024) illustrated how estimating regime probabilities instead of assigning hard labels like K-means clustering works. This inspired us to use Gaussian Mixture Models to estimate the probability of each regime and switch only when there is high conviction.

Wood, Roberts, and Zohren (2026) demonstrated the walk-forward validation method and the use of machine learning for portfolio management. We leveraged the walk-forward approach to test our models across multiple out-of-sample periods to validate robustness.

3. DATA & FEATURE ENGINEERING

3.1 DATA COLLECTION

We collected daily market data and monthly macro data from Bloomberg, and daily momentum return from the Fama-French library, spanning December 1999 to December 2025. The asset universe covers six major categories: equities, fixed income, commodities, foreign exchange, volatility, and macroeconomic indicators.

We pulled the following indices across equities, fixed income, commodities, foreign exchange, volatility, and macroeconomic indicators to construct our 49 signals for regime detection.

Category	**Tickers / Data**
Equities	SPX, RLV, RU20INTR, RTY
Fixed Income	US Treasury Yields (2Y, 10Y, 30Y), LUACTRUU, LF98TRUU, LUAS, LF98OAS
Commodities	Brent Crude Oil, GOLDLNPM
Foreign Exchange	DXY
Volatility	VIX
Macroeconomic	CPI, UMich Consumer Sentiment, Conference Board, ISM Manufacturing, ISM Services, M1, M2, NBER Recession Indicator
Factor Data	Fama-French Momentum Factor

3.2 ASSET UNIVERSE & BENCHMARK

OUR PRIMARY BENCHMARK:

Asset	**Weight**
US Equity	55%
Fixed Income	40%
Alternative	5%

ETF UNIVERSE:

We constructed our investment universe using the following liquid ETFs for the three asset classes. The investment universe used for models backtesting.

EQUITIES

Ticker	Sub-Asset Class
SPY	US Large Cap Blend
IVE	US Large Cap Value
IVW	US Large Cap Growth
IWM	US Small Cap

FIXED INCOME

Ticker	Sub-Asset Class
SHY	Short-Term Treasury (1-3Y)
IEF	Intermediate Treasury (7-10Y)
TLT	Long-Term Treasury (20Y+)
LQD	Investment Grade Corporate Bonds
HYG	High Yield Corporate Bonds

ALTERNATIVES

Ticker	Sub-Asset Class
GLD	Gold

3.3 FEATURE ENGINEERING

In order to capture all angles of the market for adequate regime identification, we constructed 49 features across twelve diverse categories. The feature buckets we used:

Bucket	**Reason**	**Features**
Equity Market Dynamics	Determine market performance	SPX return, SPX volatility, drawdown
Volatility & Risk Sentiment	Assess the fear gauge	VIX level, VIX change
Interest Rate Environment	Capture rate environment	Yield curve slope (10Y-2Y), yield change
Cross-Asset Risk Signals	Check asset correlations	Equity-bond correlation, equity-bond spread
Global Risk Conditions	Capture global stress level	DXY return, DXY momentum
Economic Activity	Assess economic activity and growth momentum	ISM Manufacturing, ISM change
Inflation Environment	Determine asset class performance across regimes	CPI YoY, CPI change, inflation regime flag
Liquidity & Monetary Conditions	Check monetary policy	M2 growth, M2 YoY
Safe Haven Signals	Capture movement to safety	Gold/SPX ratio, gold return
Commodity & Inflation Shocks	Capture sudden price shocks	Oil return, oil volatility, oil momentum
Bond Market Features	Capture bond market signals	10Y yield, 30Y yield
Credit Market Stress	Capture risk of defaults	HY spread, IG spread, HY change

A complete list of all 49 features with their definitions is provided in Appendix A.

3.4 PRINCIPAL COMPONENT ANALYSIS (PCA)

With 49 features, many of which are correlated, feeding them directly into the models would cause overfitting as the model would learn noise as a pattern due to high multicollinearity across those features. For example, features like the change in high yield spreads and the high yield spread level itself would be highly correlated. To address this, we applied Principal Component Analysis (PCA) and found that 14 components explained 80% of the variance as shows in Figure 1, which we used as inputs to our regime detection models.

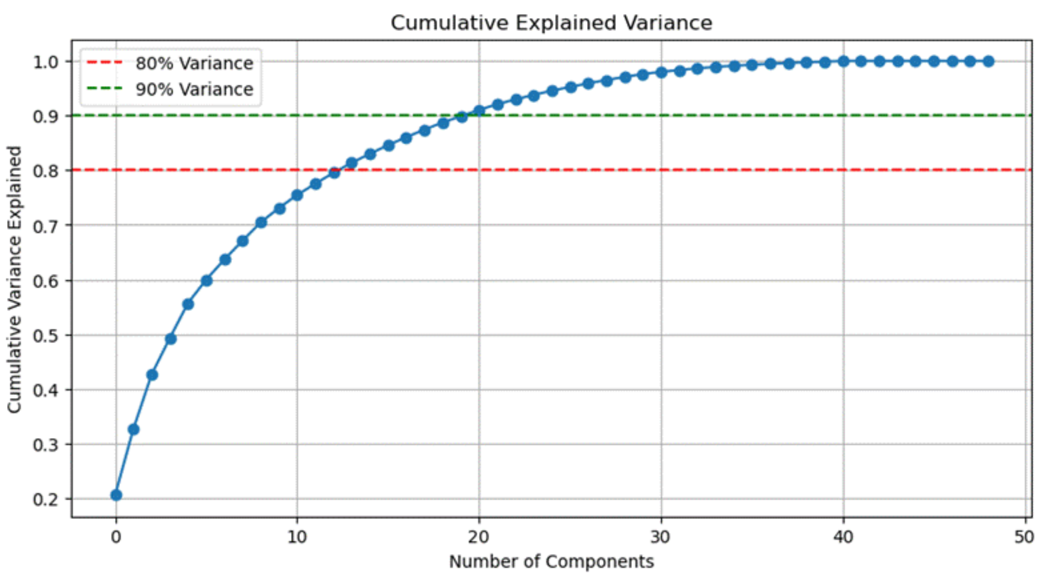


Figure 1

3.5 DETERMINING THE OPTIMAL NUMBER OF REGIMES

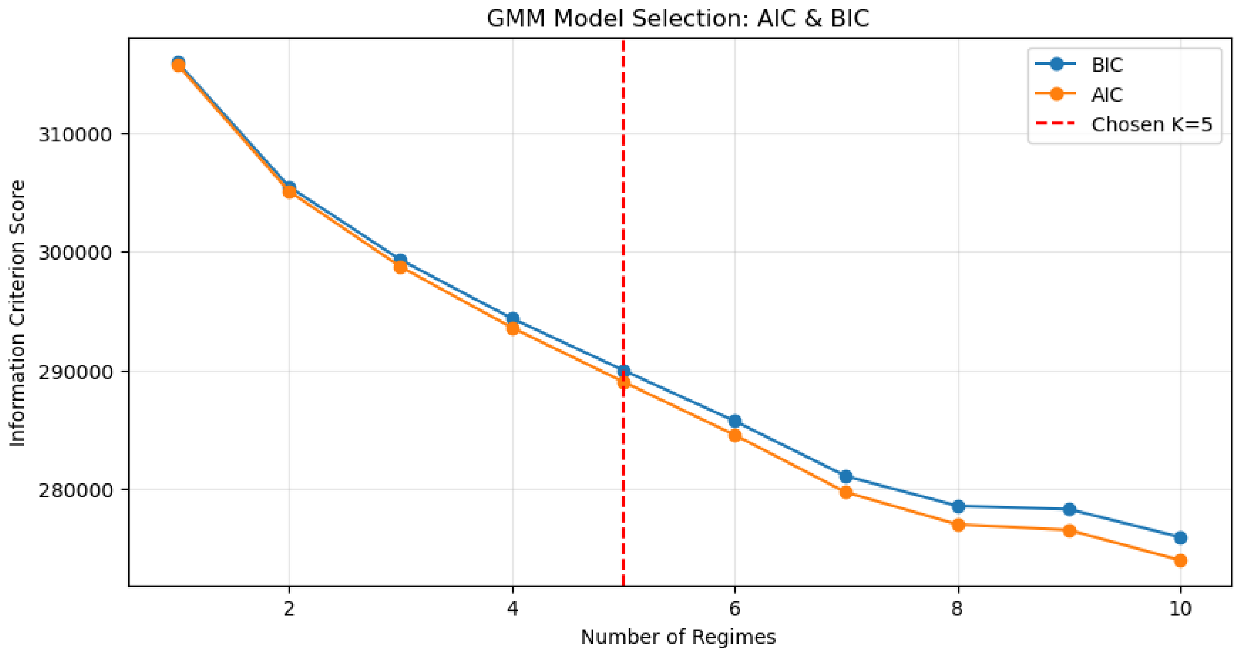


Figure 2

To determine the optimal number of regimes, we ran AIC and BIC analysis. However, AIC and BIC kept decreasing as K increased as shown in Figure 2, which did not give us a clear elbow. We then ran Silhouette analysis comparing K = 3, 4, and 5. While the Silhouette scores were not adequate across all values of K, K = 5 had the highest score among all, so we proceeded with five regimes.

Metric	**k=3**	**k=4**	**k=5**
Avg. Silhouette	0.0945	0.0665	0.1174
Model Utility	Too General	High Overlap	Optimal for Risk Signaling

3.6 REGIME IDENTIFICATION VIA GMM

We applied Gaussian Mixture Models on the full dataset from 2001 to 2025 using 14 principal components that explained 80% of the variance and K = 5.

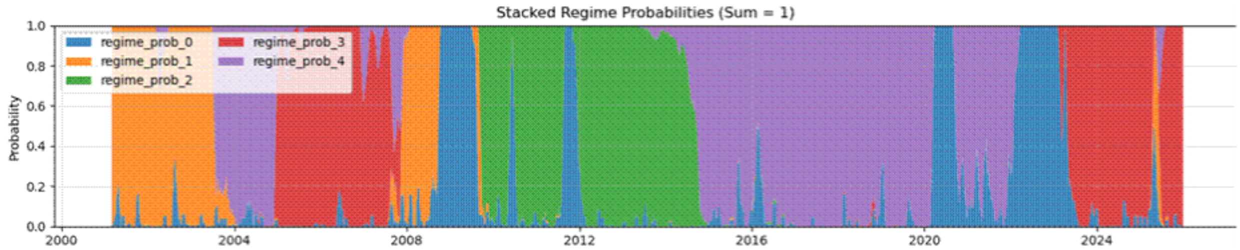


Figure 3

As shown in Figure 3, The GMM was able to identify the crash during 2008 and covid-19.

To validate the regimes, we analyzed five key features across each regime: SPX return, VIX, high yield and investment grade spreads, CPI change, and the yield curve slope between the ten-year and two-year treasury. VIX was the most discriminating feature to differentiate between regimes. This analysis identified five distinct market regimes:

Regime	Name	SPX Return	VIX	Credit Spread	Yield Curve
0	Crisis / Panic	↓ Negative	Very High (~31)	Wide	↘ Flat/Low
1	Recession / Credit Stress	↓ Negative	High (~25)	Highest	↗ Steepening
2	Expansion / Risk-On	↑ Strong Positive	Moderate (~17)	Tight	Steep
3	Late Cycle / Policy Tightening	↑ Positive	Low (~14)	Tightest	Flat
4	Goldilocks / Stable Growth	↑ Positive	Low (~16)	Tight	Normal

4. METHODOLOGY

4.1 WALK-FORWARD VALIDATION FRAMEWORK

A regular train/test split tests the model once on an out-of-sample period. In order to ensure robustness, we introduced the walk-forward method, ensuring that the model is not only working well during one test period.

The framework of the walk-forward is designed to start training the model on data from 2001 to 2018 and test on 2019. Subsequently, the model expands the training window for each fold by one year and tests on the consecutive year until it reaches 2025, giving us seven folds. To sum up, we test out of sample seven different times.

Despite having the raw data starting from 1999, the features with no missing values start from 2001. The reason we have missing values in the features is because we use rolling windows for some of the features. In order to capture all aspects of the market at the same time, we dropped all dates with missing values and ensured alignment while still capturing major market regimes including the dot-com crash, the 2008 financial crisis, and the post-crisis expansion. Additionally, the walk-forward method helps us test how fast the model learns from recent market moves without using future information.

Refitting monthly or daily might seem more ideal, however it would be expensive from a computational perspective and unnecessary for medium-long term investing. For actual investment use, we recommend refitting monthly.

4.2 ASYMMETRIC SWITCHING LOGIC

To reduce unnecessary regime switches and lower turnover, we apply an asymmetric switching logic across all models. To enter a new regime, the model requires seven consecutive days confirming the new regime with a probability of at least 75%. However, to exit a bear regime, only one day of confirmation is required. This asymmetry was designed because bear markets can experience sharp short-term rallies, and a slow exit would cause the portfolio to miss significant upside during the transition from bear to bull.

4.3 MEAN-VARIANCE PORTFOLIO CONSTRUCTION

The mean-variance optimization is a useful tool to allocate robustly across asset classes and take active positions given different regimes based on the asset return and covariance matrix for each regime. Rather than using a fixed allocation, the optimizer adjusts dynamically based on the current regime and the investor's risk tolerance.

OBJECTIVE FUNCTION

The optimizer solves the following problem at each refit:

$$maximize\ \{\ \mu^T w - \gamma\,.\,w^T \Sigma w - \lambda \left|\left|w\right|\right|^2 \}$$

where μ is the vector of expected returns, Σ is the covariance matrix, γ is the risk aversion parameter, and λ is the L2 regularization term that penalizes extreme weights.

RISK TOLERANCE & GAMMA

The optimizer is controlled by gamma, a risk aversion parameter, to determine how aggressive or conservative the allocation for each investor risk profile across regimes.

RETURN & COVARIANCE ESTIMATION

For Bull and Transition regimes, the optimizer uses the mean returns and covariance matrix estimated from the training data belonging to that regime only. For the Bear regime, we use a blended estimate to improve reliability since bear periods represent a small portion of the total data:

$$\mu_{Bear} = 0.6 \times \mu_{bearPeriods} + 0.4 \times \mu_{FullHistory}$$

$$\Sigma_{Bear} = 0.6 \times \Sigma_{bearPeriods} + 0.4 \times \Sigma_{FullHistory}$$

If any regime has fewer than 20 observations in the training period, the optimizer falls back to the full training dataset to ensure the estimates passed to the optimizer are reliable.

REGIME-SPECIFIC CONSTRAINTS

Constraint	**Bull**	**Transition**	**Bear**
Equity Min/Max	60% / 80%	45% / 65%	Unconstrained
Bond Min/Max	10% / 40%	20% / 45%	Min floor per risk profile
Gold Max	20%	25%	20%
Active Tilt Limit	35%	25%	20%
Group Tilt Limit	40%	30%	25%

In the Bear regime, equity allocation is left unconstrained with a minimum bond floor, for example, 40% for Conservative and 25% for Moderate, is maintained based on the investor's risk profile, to ensure capital preservation during stress periods.

INTEREST RATE ADJUSTMENT

During periods of rising interest rates, long-term bonds such as TLT can suffer significant losses. We noticed the optimizer would sometimes allocate heavily to TLT in rising rate environments. To address this, when both a Bear regime and rising yields are detected simultaneously, we cap TLT and IEF allocations at 5% each and raise the bond minimum to 35% to shift toward shorter duration bonds. This minimizes portfolio tail risk during the worst-case scenario of a simultaneous equity and rate shock, which is exactly what happened in 2022.

ADDITIONAL CONSTRAINTS

To prevent concentration, we applied a maximum weight of 40% per ETF and an L2 regularization term of 0.00005 to penalize large deviations.

MV BENCHMARK

As one of our two benchmarks, we also construct a static mean-variance portfolio using the same walk-forward structure but without any regime conditioning. This benchmark uses a fixed gamma of 2.0 and applies the same constraints across all periods, allowing us to isolate the value added by regime detection specifically.

4.4 MERGING 5 REGIMES TO 3

Rather than using the five GMM regimes directly to train XGBoost, we merged them into three regimes (Bull, Transition, and Bear). Using rule-based labels derived from VIX and SPX drawdown. Predicting five regimes directly would reduce prediction accuracy since some regimes have very short periods, leaving the model with insufficient data to learn about the regime. By merging to three regimes, the model has higher prediction accuracy and more observations per regime to train on as shown in XGBoost confusion matrices for 5 regimes and 3 regimes prediction in Figure 4. Based on the data analysis, VIX and SPX drawdown were the most discriminator features. The thresholds used are:

- SPX drawdown below -12.5% or VIX above 25: Bear
- VIX above 18: Transition

- Otherwise: Bull

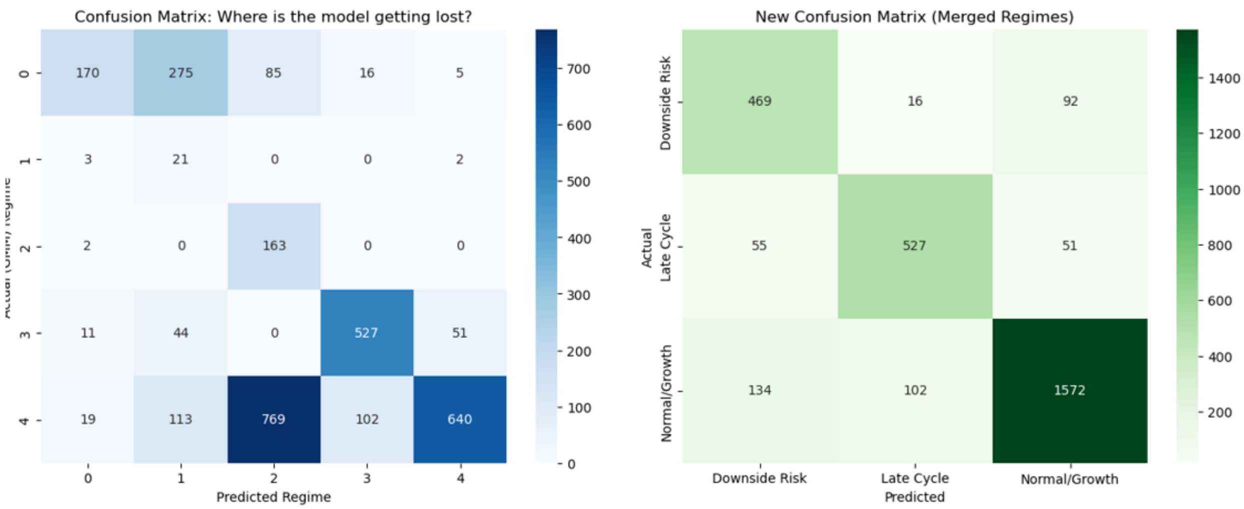


Figure 4

4.5 ML MODEL (GMM + XGBOOST)

We used a hybrid method for the machine learning model. The model combines unsupervised learning, GMM, for regime identification and a supervised classifier, XGBoost, for regime prediction.

The model splits the data into training and testing periods following the walk-forward validation framework described in Section 4.1.

On top of the 49 base features, we added three augmented discriminator features: the 63-day yield trend, the 21-day cumulative SPX return, and the interaction between VIX and the yield trend. These features were added during the testing phase and showed promising results in improving regime prediction alongside the existing features.

All features are standardized before applying PCA. We use 14 principal components, which explain 80% of the variance, as inputs to the GMM. The GMM captures the latent structure of each regime using the training data only.

Rather than using GMM output directly to train XGBoost, we merge the five regimes into three as described in Section 4.4. XGBoost is then trained on all features, to predict the three rule-based regime labels.

We initially tested LSTM, but it did not perform well due to the limited length of our time series. We also tested Random Forest, but XGBoost achieved higher prediction accuracy and was selected as the final classifier.

To reduce unnecessary regime switches, we apply the asymmetric switching logic described in Section 4.2.

4.6 HIDDEN MARKOV MODEL (HMM)

4.6.1 OVERVIEW

It is structured as an end-to-end quantitative pipeline that transforms raw market and macroeconomic data into dynamically rebalanced multi-asset portfolio weights. It comprises four principal components: (i) feature engineering and data preparation, (ii) unsupervised latent-state identification via Gaussian Hidden Markov Models (HMMs), (iii) regime-conditional Mean-Variance Optimization (MVO), and (iv) a walk-forward out-of-sample backtest with monthly rebalancing. Each component is described in detail in the subsections that follow.

The full pipeline includes data ingestion, feature engineering, HMM regime identification, forward return validation, MVO weight optimization, and walk-forward backtesting is implemented in Python and evaluated across four strategy variants (K ∈ {3, 5} × objective ∈ {Max-Sharpe, Min-Vol}). This report provides a comprehensive technical account of every component.

4.6.2 PREPROCESSING PIPELINE

Three preprocessing steps are applied before HMM estimation:

1. Forward Fill: Missing values (especially lagged macro indicators) are propagated forward from the last valid observation, avoiding look-ahead bias.
2. Winsorization: Each feature is clipped to its 1st/99th percentile on training data, bounding crisis-period outliers without discarding them.
3. Standardization: A zero-mean, unit-variance scaler is fitted on the training window and applied to both train and test data, preventing high-magnitude features (e.g. VIX) from dominating the HMM likelihood.

4.6.3 MODEL SPECIFICATION

A Gaussian HMM models a latent state sequence $s_1, s_2 \ldots, s_T$ *driving a multivariate observation sequence* $x_1, x_2 \ldots, x_T$. The model is parameterized by π (initial distribution), A ($K \times K$ transition matrix), and per-state emission $\{\mu_k, \Sigma_k\}$. Emissions follow multivariate Gaussians with full covariance matrices, capturing the rich cross-feature dependencies that characterize each regime:

$$p(x\_t \mid s\_t = k) = N(x\_t ; \mu\_k, \sigma\_k)$$

4.6.4 PARAMETER ESTIMATION VIA BAUM-WELCH (EM)

HMM parameters are estimated by maximizing the marginal log-likelihood of the observation sequence:

$$theta* = \arg\ \max \_\{theta\}\ \log P\ (x_{1:T} \mid theta|)$$

This is iterating between the forward-backward E-step (computing posterior state occupancies $\gamma_{t(k)}$ and transition posteriors $\xi\_t(i,j))$ and the M-step (closed-form updates for μ_k, Σ_k, and A). Convergence tolerance is 1×10^{-5}; maximum iterations are 2,000. To escape local optima, 30 random initializations are used in full-sample estimation and 10 per walk-forward window, retaining the highest log-likelihood solution.

4.6.5 CANONICAL STATE ORDERING

The label-switching problem is resolved by ordering states in ascending HY spread mean — the most direct single-dimensional proxy for credit risk appetite so that State 0 always represents Risk-On and State K-1 always represents Risk-Off/Crisis.

$$order = \arg s\,ort\left(\mu_{\{k,\, HY_{Spread}\}}\right), \quad k = 0,\ 1,\ \dots,\ K-1$$

Two resolution configurations are estimated: $K = 3$ (Risk-On / Transitional / Risk-Off) and $K = 5$ (Risk-On / Mild Growth / Transitional / Mild Stress / Risk-Off/Crisis). Finally, a rolling 10-observation modal filter suppresses economically implausible single-day state switches in the decoded sequence.

4.6.6 REGIME CONFIGURATIONS: K = 3 AND K = 5

Two resolution configurations are estimated in parallel. Table 4.6.1 and Table 4.6.2 describe the resulting state taxonomies.

State	Label	Macroeconomic Characteristics	Portfolio Implication

0	Risk-On	Low HY spreads, positive SPX momentum, compressed VIX, steep yield curve, constructive ISM	High equity (30-65%)
1	Transitional	Mixed signals: moderate VIX, flat/ambiguous yield curve, unclear credit direction	Balanced; gold floor 8%
2	Risk-Off	Elevated HY spreads, negative SPX return, high VIX, weak sentiment, flat/inverted curve	Defensive: equity <=20%, bonds+gold+cash >=60%

Table 4.6.1: Three-state (K = 3) regime taxonomy

State	Label	Characteristics	Equity Bounds
0	Risk-On	Strong growth, tight credit, positive SPX momentum, low VIX, steep curve	40-70%
1	Mild Growth	Moderating but positive signals; stable credit; VIX low-to-moderate	30-60%
2	Transitional	Ambiguous cross-asset signals; flat yield curve; indeterminate credit	15-45%
3	Mild Stress	Widening credit spreads, rising VIX, slightly negative SPX drift, weakening sentiment	8-30%
4	Risk-Off / Crisis	Acute stress: high VIX, wide HY spreads, flight-to-safety, elevated gold premium	2-15%

Table 4.6.2: Five-state (K = 5) regime taxonomy

4.7 HIDDEN SEMI-MARKOV MODEL (HSMM)

The Hidden Semi-Markov Model (HSMM) was implemented as one of the primary regime-detection models within the RAAM framework. The objective of the model is to identify hidden market regimes and determine how financial markets transition between different economic environments over time. Unlike a traditional Hidden Markov Model (HMM), which assumes that state transitions occur randomly with fixed probabilities, the HSMM explicitly models how long a regime tends to persist before switching into another regime. This is important because financial markets usually remain within the same economic environment for extended periods rather than changing states every few days.

The HSMM assumes that market regimes are latent and cannot be directly observed. Instead, the model infers the current regime using observable financial and macroeconomic variables. By analyzing patterns across volatility, credit conditions, equity performance, inflation, and interest rates, the HSMM estimates the most probable hidden state driving market behavior at each point in time.

The methodology consists of four main stages:

1. Feature preprocessing and dimensionality reduction
2. Initial latent-state identification
3. Duration and transition estimation
4. Regime decoding and classification

4.7.1 FEATURE PREPARATION AND PCA

The HSMM framework begins by constructing a feature matrix containing financial and macroeconomic indicators designed to capture multiple dimensions of market behavior. The variables include equity returns, VIX levels, Treasury yield spreads, credit spreads, inflation indicators, ISM data, dollar index momentum, and gold returns.

Because many financial variables are highly correlated, directly feeding all variables into the HSMM can lead to unstable estimation and overfitting. To reduce dimensionality, Principal Component Analysis (PCA) is applied after standardizing the data.

PCA transforms the original feature space into a smaller set of orthogonal principal components that preserve most of the information contained in the dataset. This allows the HSMM to work with a cleaner lower-dimensional representation of the market environment while reducing noise and multicollinearity.

4.7.2 INITIAL REGIME IDENTIFICATION

Before estimating the full HSMM, the framework initializes latent states using K-Means clustering within the PCA-transformed feature space. K-Means groups observations with similar financial and macroeconomic characteristics into the same cluster.

The purpose of this initialization step is to provide a preliminary estimate of the hidden market regimes before estimating the probabilistic HSMM structure. Multiple state configurations were evaluated, and the five-regime framework produced the clearest economic separation between market environments.

The identified regimes were later interpreted economically as:

- Crisis / Panic
- Recession / Credit Stress
- Expansion / Risk-On
- Late Cycle / Policy Tightening
- Goldilocks / Stable Growth

4.7.3 EMISSION DISTRIBUTION ESTIMATION

After initializing the latent states, the HSMM estimates the statistical characteristics of each regime using Gaussian emission distributions.

An emission distribution defines the probability of observing a particular feature vector conditional on being in a specific hidden state. For each regime, the model estimates:

- Mean vector
- Covariance matrix

The mean vector captures the average behavior of variables inside a regime, while the covariance matrix captures the relationships between variables during that regime.

For example, crisis regimes are generally associated with elevated volatility, negative equity returns, and widening credit spreads, while expansionary regimes tend to exhibit positive market momentum and lower volatility conditions.

These estimated distributions allow the model to compare new observations against the learned characteristics of each regime and determine which hidden state is most likely active.

4.7.4 DURATION MODELING

The main difference between a standard HMM and an HSMM is the explicit modeling of regime duration.

Traditional HMMs assume that the probability of leaving a state is constant at every time step, which can produce excessive short-term switching. Financial markets, however, usually remain in the same economic environment for sustained periods.

To address this issue, the HSMM directly estimates how long each regime tends to persist before transitioning into another state.

The duration estimation process consists of:

1. Recording consecutive runs of each regime
2. Measuring the length of each regime sequence
3. Estimating duration probabilities for each state

These duration probabilities are incorporated into the regime-detection process. If a regime historically persists for longer periods, the model assigns lower probability to immediate short-term transitions. This helps filter temporary market noise and produces smoother regime classifications.

4.7.5 TRANSITION MATRIX ESTIMATION

After estimating regime durations, the HSMM estimates transition probabilities between regimes.

The transition matrix describes the probability of moving from one regime into another once the expected duration of the current regime has ended. Unlike a standard HMM, persistence is modeled separately through duration distributions rather than repeated self-transitions.

The transition matrix allows the model to learn realistic macroeconomic progression patterns. For example, expansionary periods may transition into late-cycle tightening environments, while crisis regimes may evolve into recessionary conditions before recovery occurs.

4.7.6 REGIME DETECTION USING VITERBI DECODING

After estimating the emission distributions, duration probabilities, and transition matrix, the HSMM applies an extended Viterbi decoding algorithm to determine the most likely sequence of market regimes.

The decoding process is the core mechanism that allows the model to detect regimes over time.

At each time step, the algorithm evaluates:

- The likelihood of the observed data under each regime
- The probability of remaining in a regime for a given duration
- The probability of transitioning between regimes

The algorithm recursively compares all possible state and duration combinations to identify the most probable path through time.

The decoding process proceeds through the following steps:

1. Evaluate candidate regimes
2. Compute observation likelihoods
3. Apply duration probabilities
4. Apply transition probabilities
5. Store the optimal path dynamically

At the end of the sequence, the algorithm reconstructs the most likely historical regime path by backtracking through the stored probabilities.

This allows the HSMM to detect market regimes not only by matching market characteristics, but also by considering how long regimes typically persist and how economic environments usually evolve over time.

4.7.7 ECONOMIC INTERPRETATION OF REGIMES

After decoding the optimal regime sequence, each latent state is interpreted economically using the characteristics observed within that state.

The resulting regimes are classified as:

Regime	**Economic Interpretation**
0	Crisis / Panic
1	Recession / Credit Stress
2	Expansion / Risk-On
3	Late Cycle / Policy Tightening
4	Goldilocks / Stable Growth

This converts the latent statistical states into economically interpretable market environments that can later be integrated into the portfolio allocation framework.

Overall, the HSMM provides a probabilistic framework for detecting persistent market regimes through the interaction of observable financial variables, duration modeling, and transition dynamics. By explicitly modeling regime persistence, the methodology produces more stable and economically realistic regime classifications than traditional Markov models.

4.8 CONTINUOUS JUMP MODEL (CJM)

4.8.1 PURPOSE OF THE JUMP MODEL

The jump model is developed to capture market behavior that cannot be explained by smooth volatility alone. Financial returns often show two distinct dynamics: (i) continuous small fluctuations during normal market conditions, and (ii) sudden large discontinuous movements caused by news shocks, liquidity breaks, or macro events. A standard jump model captures the first dynamic but underestimates the second. The continuous jump model framework addresses this by combining both components in one stochastic process, improving risk measurement, and drawdown control in portfolio decisions.

4.8.2 MODEL FORMULATION

Let S_t be the asset price and $X_t = \ln(S_t)$ be the log-price. The jump diffusion process is:

$$dX_t = (\mu - ½\,\sigma^2)dt + \sigma\, dW_t + J_t\, dN_t$$

where μ is drift, σ is continuous volatility, W_t is a standard Brownian motion, N_t is a Poisson jump-count process with intensity lambda, and J_t is jump magnitude.

In discrete form over interval Δt:

$\Delta X_t = \mu \Delta t + \sigma \surd (\Delta t)\, \varepsilon_t + \sum i{=}1Yi$

with $\varepsilon_t \sim N(0,1)$, $N_t \sim \text{Poisson}(\lambda \Delta t)$, and Y_i as jump sizes (commonly $Y_i \sim N(\mu_J, \sigma_J^2)$).

4.8.3 DATA PREPARATION

The model is estimated on cleaned daily log returns:

$$r_t = \ln\left(\frac{S_t}{S_{t-1}}\right)$$

Preparation steps include aligning time series, handling missing observations, removing data errors, and computing rolling volatility. This preserves true tail events required for jump estimation.

4.8.4 JUMP IDENTIFICATION

Before full maximum-likelihood estimation, price jumps are identified using a volatility-scaled threshold:

$|r_t| > k\sigma$

where σ_t is rolling volatility and k is a threshold parameter. This preliminary label is used to initialize λ, μ_J, and σ_J for stable optimization.

4.8.5 PARAMETER ESTIMATION

The parameter vector is:

$\Theta = \{\mu, \sigma, \lambda, \mu_J, \sigma_J\}$

Parameters are estimated with Maximum Likelihood Estimation (MLE). The return density is represented as a mixture across possible jump counts:

$$p(rt) = \sum_{n=0}^{n_{max}} P(Nt = n)\ f\left(\frac{r_t}{n_{jumps}}\right)$$

where n_{max} is a practical truncation level (usually small for daily data). To handle market non-stationarity, estimation is run in rolling windows so jump intensity and jump severity can adapt over time.

4.8.6 JUMP-RISK SIGNAL CONSTRUCTION

From estimated parameters, a jump-risk score is computed:

$$JumpRisk_t = \lambda\hat{t}(\hat{\mu}^2 J + \hat{\sigma}^2 J)$$

Higher $JumpRiskt$ implies greater discontinuity risk in the next allocation period.

4.8.7 INTEGRATION WITH PORTFOLIO CONSTRUCTION

The continuous jump module is integrated into portfolio decisions through risk-aware exposure control: reducing position sizes when $JumpRiskt$ is high, penalizing assets with persistent negative jump behavior, and tightening constraints during high jump-risk regimes.

4.8.8 VALIDATION AND ROBUSTNESS

Robustness checks include varying the threshold k, rolling window length, and jump-size distribution assumptions, as well as re-estimation across market regimes.

4.8.9 METHODOLOGICAL CONTRIBUTION

This continuous jump model methodology extends conventional volatility modeling by incorporating discontinuous shocks. It improves realism in return dynamics, strengthens tail-risk management, and enhances regime-aware allocation decisions within the overall portfolio framework.

5. BACK TEST ENGINE

5.1 DAILY RETURN SIMULATION

The back test engine calculates the time series of daily portfolio returns using the weights assigned by each model. The weights are determined by the mean-variance optimizer output for each detected regime. For the benchmarks, the weights are either fixed at the static benchmark allocation or determined by the MV no-regime optimizer.

The daily portfolio return is calculated as:

$$r_{p,t} = w^T \times r_t - TC_t$$

where w is the vector of portfolio weights, r_t is the vector of daily ETF returns, and TC_t is the transaction cost incurred on rebalancing days. Transaction costs are assumed to be 5 basis points and are applied each time the portfolio changes allocation.

A minimum holding period is applied per regime to limit unnecessary switching and reduce turnover:

Regime	**Minimum Holding Period**
Bull	21 days
Transition	15 days
Bear	10 days

The Bear regime has the shortest minimum holding period because we noticed that some rallies may happen after a bear market. To participate in the rally, we assigned the shortest holding period to Bear.

5.2 BASELINES

We compare all models against 2 baselines. The first is a static benchmark (55% equity, 40% bonds, and 5% gold) rebalanced semi-annually. The second is a MV no-regime benchmark using the same walk-forward structure but without any regime conditioning, using a fixed gamma of 2.0. This benchmark helps determining the value added from the regime.

5.3 PERFORMANCE METRICS

We evaluate all models using the following metrics:

Metric	**Formula**
Annual Return	$(1+r)^{\frac{252}{n}} - 1$
Annual Volatility	$\sigma \times \sqrt{252}$
Sharpe Ratio	$\frac{(\mu \times 252)}{(\sigma \times \sqrt{252})}$
Sortino Ratio	$\frac{(\mu \times 252)}{(\sigma_{down} \times \sqrt{252})}$
Calmar Ratio	$\frac{Annual\ Return}{Abs(Max\ Drawdown)}$
Max Drawdown	$\min\left(\frac{Cum_t - Cum_{max}}{Cum_{max}}\right)$
Win Rate	% of months with positive return
Switches	Total number of regime changes

5.4 REGIME ATTRIBUTION

The regime attribution table breaks down the model performance by each detected regime: Bull, Transition, and Bear. We calculated:

- annualized return
- benchmark return
- alpha

- Sharpe ratio
- delta Sharpe versus the benchmark
- Sortino ratio
- maximum drawdown.

This allows us to attribute the performance for each regime and check if the model was useful to add value through each state.

6. RESULTS

6.1 OVERALL PERFORMANCE FOR ALL MODELS VS BENCHMARK

The Jump and Machine Learning models were the highest performing models compared to HMM and HSMM. Both models achieved approximately the same Sharpe ratio of 1.07, which is the highest across all models and benchmarks. The Jump model had the highest annual return at 13.78%, while the Machine Learning model maintained the same volatility as the benchmark at 12.44% with better maximum drawdown control at -24.8% compared to the Jump model's -29.7%.

Metric	**Static BM**	**MV No-Regime**	**ML**	**Jump**	**HMM**	**HSMM**
Final Value	$175,379	$206,240	$239,406	$246,396	$207,187	$208,289
Ann. Return	8.38%	10.92%	13.31%	13.78%	10.99%	11.08%
Ann. Vol	12.43%	11.88%	12.44%	12.82%	13.15%	14.72%
Sharpe	0.710	0.932	1.067	1.072	0.859	0.788
Sortino	0.664	0.895	0.998	1.016	0.800	0.737
Calmar	0.374	0.409	0.536	0.464	0.385	0.394
Max Drawdown	-22.4%	-26.7%	-24.8%	-29.7%	-28.5%	-28.1%
Win Rate	66.7%	65.5%	66.7%	70.2%	64.3%	65.5%
Switches	0	0	34	8	31	25

Interestingly, the Jump model achieved this high Sharpe ratio with only 8 switches compared to the ML model that switched 34 times, leading to significantly lower transaction costs. The key trade-off between the 2 models is higher turnover with better drawdown control for the ML model versus lower turnover with higher volatility for the Jump model.

6.2 ANNUAL PERFORMANCE BREAKDOWN

Year	Static BM	MV	ML	Jump	HMM	HSMM
2019	19.2%	18.2%	22.8%	22.4%	21.9%	20.5%
2020	14.7%	21.3%	19.2%	31.0%	28.0%	23.6%
2021	11.1%	5.4%	19.9%	17.7%	11.1%	10.1%
2022	-17.9%	-20.8%	-20.9%	-25.2%	-24.7%	-21.9%
2023	13.6%	13.3%	20.7%	15.6%	12.2%	15.5%
2024	10.3%	17.8%	23.7%	20.6%	17.2%	16.0%
2025	12.3%	29.1%	15.6%	25.1%	20.7%	21.3%

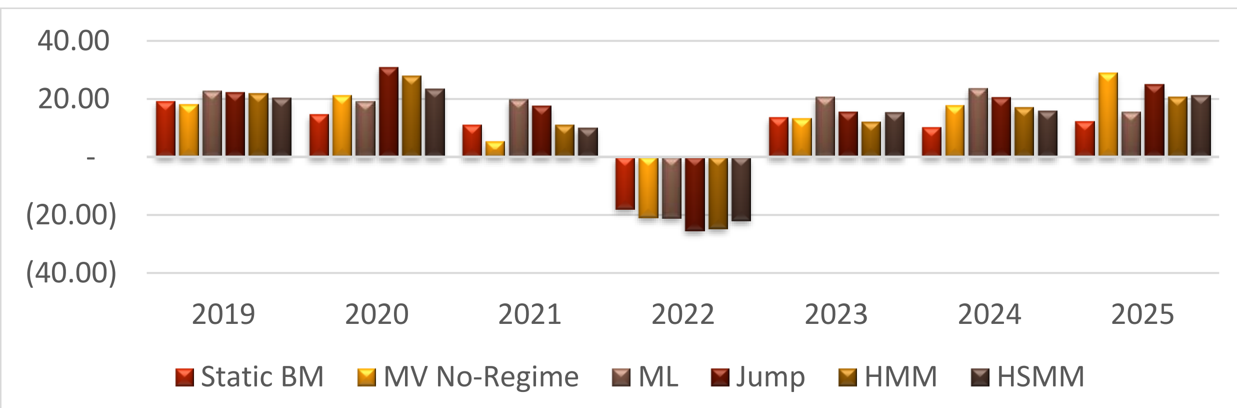


Figure 5

Both the ML and Jump models outperformed the static benchmark in 6 out of 7 years as shown in Figure 5. The exception was 2022, where all models underperformed. The ML model's more frequent switching helped it limit losses to -20.9% in 2022 compared to the Jump model's -25.2%.

6.3 REGIME ALPHA ATTRIBUTION

ML Model | Regime Alpha:

Regime	Days	% Time	Model Ret	BM Ret	Alpha	Model Sharpe	BM Sharpe	Delta Sharpe	Sortino
Bull	988	56.1%	15.72%	8.20%	+7.52%	1.328	1.000	+0.328	1.242
Transition	467	26.5%	8.50%	4.48%	+4.02%	0.704	0.420	+0.284	0.663
Bear	305	17.3%	13.15%	15.24%	-2.09%	0.910	0.777	+0.132	0.844

The ML model generated positive alpha in both Bull and Transition regimes at +7.52% and +4.02% respectively. In the Bear regime, the model had slightly negative alpha of -2.09%, however the Sharpe ratio remained above the benchmark across all regimes indicating better risk-adjusted performance even during stress periods.

Jump Model | Regime Alpha:

Regime	Days	% Time	Model Ret	BM Ret	Alpha	Model Sharpe	BM Sharpe	Delta Sharpe	Sortino
Bull	310	17.6%	22.86%	13.83%	+9.03%	1.385	1.429	-0.044	1.330
Transition	1342	76.2%	10.93%	5.03%	+5.90%	0.950	0.499	+0.451	0.915
Bear	108	6.1%	25.20%	39.03%	-13.8%	1.409	1.342	+0.067	1.180

The Jump model spends 76.2% of the time in Transition, which explains its low turnover and stable allocation. It generated strong alpha in both Bull and Transition regimes. The significant negative alpha in Bear at -13.83% is explained by the model missing the recovery due to its slower switching logic.

6.4 RISK METRICS

Metric	Static BM	MV No-Regime	ML	Jump	HMM	HSMM
Ann. Vol	12.43%	11.88%	12.44%	12.82%	13.15%	14.72%
VaR (95%)	-1.124%	-1.140%	-1.248%	-1.255%	-1.337%	-1.388%
CVaR (95%)	-1.846%	-1.741%	-1.865%	-1.907%	-2.036%	-2.200%
Skewness	-0.418	-0.063	-0.309	-0.112	-0.300	-0.266
Excess Kurtosis	12.000	6.362	3.033	5.097	3.922	7.343
Max Drawdown	-22.4%	-26.7%	-24.8%	-29.7%	-28.5%	-28.1%

The Static Benchmark has the highest excess kurtosis at 12.0, indicating heavier tails and more extreme daily moves. The ML model has the lowest excess kurtosis at 3.033, closest to a normal distribution across all models, indicating more stable daily returns. The Jump model has better skewness at -0.112 compared to ML at -0.309, meaning fewer extreme negative days. HSMM has the highest volatility and worst VaR/CVaR across all models.

6.5 TURNOVER & TRANSACTION COSTS

A transaction cost of 5 basis points per trade was applied each time the portfolio changed allocation.

All Models Turnover Summary:

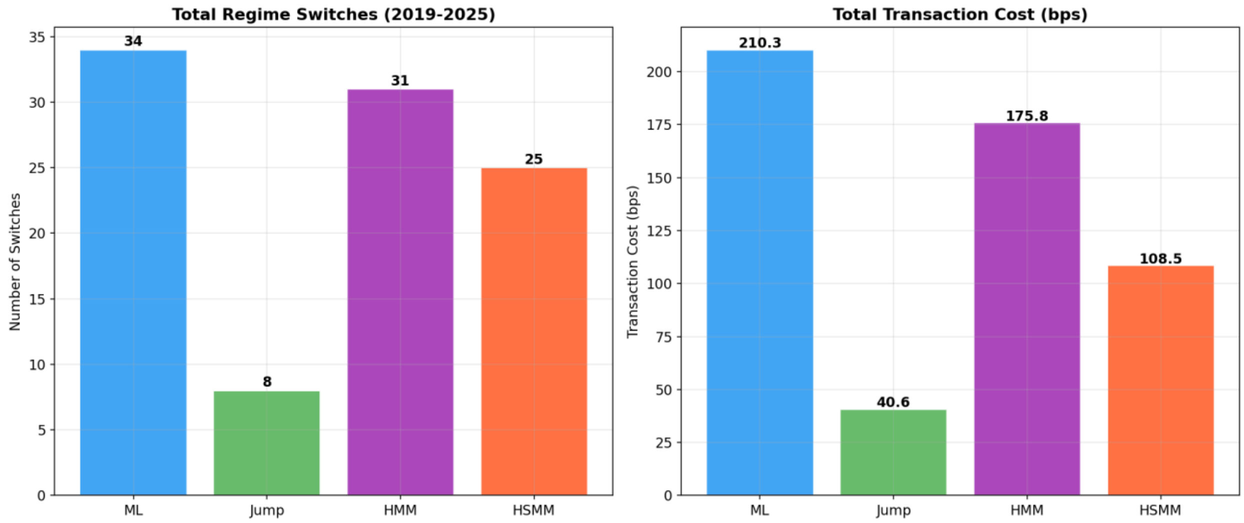


Figure 6

Model	Total Switches	Total Turnover	Total TC (bps)
ML	34	42.025	210.3
Jump	8	8.106	40.6
HMM	31	35.142	175.8
HSMM	25	21.703	108.5

ML Model | Annual Turnover:

Year	Switches	Total Turnover	Avg/Switch	TC (bps)
2019	4	4.750	1.188	23.8
2020	2	2.211	1.105	11.1
2021	6	8.056	1.343	40.3
2022	8	7.470	0.934	37.4

Year	Switches	Total Turnover	Avg/Switch	TC (bps)
2023	4	4.553	1.138	22.8
2024	4	6.763	1.691	33.8
2025	6	8.222	1.370	41.1
TOTAL	34	42.025	1.236	210.3

Jump Model | Annual Turnover:

Year	Switches	Total Turnover	Avg/Switch	TC (bps)
2019	1	1.000	1.000	5.0
2020	2	2.339	1.169	11.7
2021	1	1.092	1.092	5.5
2022	2	2.184	1.092	10.9
2023	0	0.000	0.000	0.0
2024	0	0.000	0.000	0.0
2025	2	1.491	0.746	7.5
TOTAL	8	8.106	1.013	40.6

The Jump model switched only 8 times over the full 7-year period compared to 34 times for the ML model, resulting in significantly lower total transaction costs of 40.6 bps versus 210.3 bps. The ML model switched most frequently in 2022 with 8 switches as it attempted to react to the regime shift faster. The Jump model had 0 switches in both 2023 and 2024, maintaining a fully stable allocation for 2 consecutive years.

6.6 DRAWDOWN ANALYSIS

COVID-19 Drawdown:

Model	Peak	Trough	Recovery	Depth	DD Days	Rec Days
Static BM	2/20/2020	3/23/2020	7/21/2020	-22.1%	32	120
MV	2/21/2020	3/18/2020	5/27/2020	-16.9%	26	70
ML	2/19/2020	3/19/2020	7/22/2020	-18.4%	29	125
Jump	2/19/2020	3/19/2020	7/15/2020	-16.9%	29	118
HMM	2/19/2020	3/20/2020	5/26/2020	-17.3%	30	67
HSMM	2/19/2020	3/18/2020	7/15/2020	-24.0%	28	119

- Jump had the shallowest drawdown at -16.9%, providing the best protection during the crash
- HMM recovered the fastest at 67 days
- Static BM had the deepest drawdown at -22.1% with no regime adaptation
- HSMM was the worst performer with -24.0%, deeper than the static benchmark

2022 Rate Shock Drawdown:

Model	Peak	Trough	Recovery	Depth	DD Days	Rec Days
Static BM	11/8/2021	10/14/2022	7/15/2024	-22.4%	340	640
MV	11/9/2021	10/20/2022	7/16/2024	-26.7%	345	635

Model	Peak	Trough	Recovery	Depth	DD Days	Rec Days
ML	11/19/2021	10/20/2022	2/9/2024	-24.8%	335	477
Jump	1/3/2022	11/3/2022	7/16/2024	-29.7%	304	621
HMM	11/8/2021	10/14/2022	11/7/2024	-28.5%	340	755
HSMM	11/8/2021	10/14/2022	7/16/2024	-28.1%	340	641

- ML had the shallowest drawdown at -24.8% and the fastest recovery at 477 days
- Static BM took 640 days to recover, nearly double ML's recovery time
- Jump had the deepest drawdown at -29.7% but still recovered before the static benchmark
- HMM had the longest recovery at 755 days, the worst performance during this period

6.7 MODEL COMPARISON SUMMARY

Metric	Static BM	MV	ML	Jump	HMM	HSMM
Ann. Return	8.38%	10.92%	13.31%	13.78%	10.99%	11.08%
Sharpe	0.710	0.932	1.067	1.072	0.859	0.788
Sortino	0.664	0.895	0.998	1.016	0.800	0.737
Calmar	0.374	0.409	0.536	0.464	0.385	0.394
Max Drawdown	-22.4%	-26.7%	-24.8%	-29.7%	-28.5%	-28.1%
VaR (95%)	-1.124%	-1.140%	-1.248%	-1.255%	-1.337%	-1.388%
CVaR (95%)	-1.846%	-1.741%	-1.865%	-1.907%	-2.036%	-2.200%

Metric	**Static BM**	**MV**	**ML**	**Jump**	**HMM**	**HSMM**
Win Rate	66.7%	65.5%	66.7%	70.2%	64.3%	65.5%
Switches	0	0	34	8	31	25
Total TC (bps)	--	--	210.3	40.6	175.8	108.5
Overall Rank	6th	4th	1st	1st	5th	3rd

ML and Jump are both ranked 1st as they achieved the same Sharpe ratio of approximately 1.07. ML leads with drawdown control and return distribution while Jump leads with turnover efficiency and annual return.

7. DASHBOARD & IMPLEMENTATION: RAAM

7.1 OVERVIEW & ARCHITECTURE

The RAAM dashboard is built using Shiny for Python, providing an interactive interface that allows investors to personalize their investment experience and interact with the models.

The dashboard allows the investor to try different risk tolerances, different models, and different benchmarks beyond the standard 55/40/5 allocation. This helps to validate the robustness of the models.

The overall flow of the dashboard is designed to guide the investor from start to finish:

- Starting with a risk profiling survey to determine their risk tolerance
- Running their chosen model
- Exploring the results through multiple analytical tabs covering performance, risk metrics, regime attribution, turnover, portfolio weights, and model comparison
- Exporting a personalized client report as a PDF

7.2 RISK PROFILING MODULE

The dashboard starts with a welcome page where the investor enters their name, age, and investment amount before being directed to the risk profiling survey.

To determine the investor's risk profile, which will be used to calculate gamma for regime-based mean-variance optimization and suggest the appropriate benchmark, the investor is asked to fill out an 8-question survey. The questions are designed to capture the investor's investment horizon, risk appetite, and risk tolerance. A complete list of questions and answers is provided in Appendix B.

The survey scoring methodology works as follows:

We first use Q4, “Which investment feels right for you?”, to determine the base score for each user, where Very Conservative, Conservative, Moderate, and Aggressive are assigned scores of 6, 4, 2.5, and 1.5, respectively. Subsequently, we assess the user’s risk appetite using Q2, “Your portfolio drops 25% in 3 months. What do you do?”, Q5, “Which would upset you more?”, and Q3, “Most you could lose and stay invested?”. Finally, we capture the user’s ability to tolerate risk through their investment horizon, Q1, income stability, Q6, and age.

Question	Answer	Adjustment / Constraint
Q2	Sell everything	+1.5
	Sell some	+0.5

	Hold	0.0
	Buy more	-0.5
Q5	Missing gains	+0.8
	Neutral	0.0
	Taking losses	-0.5
Q6	Very unstable	+0.8
	Somewhat unstable	+0.3
	Stable	0.0
	Very stable	-0.3
Age	35–50	+0.3
	Other	0.0
Q1	< 2 years	Floor = 5.0
	2–5 years	Floor = 3.5
	5–10 years	Floor = 2.0
	10+ years	Floor = 1.0
Q3	< 5%	Ceiling = 8.0
	≤ 15%	Ceiling = 5.5
	≤ 30%	Ceiling = 3.5
	> 30%	Ceiling = 2.0

Raw Score Equation:

$$\gamma_{raw} = min(max(Base + Adjustments, Floor), Ceiling)$$

Knowledge & Confidence Blend (Q7, Q8):

We lastly capture the user's knowledge and confidence using Q7, "How familiar are you with investing?", and Q8, "How confident are you in the answers you just gave?". The knowledge and confidence scores are used to determine K where $K = \frac{Q_7+Q_8}{2}$. Finally, we use K and raw gamma value to calculate the final gamma.

$$\gamma_{final} = K \times \gamma_{raw} + (1 - K) \times 3.5$$

Profile Assignment:

Once gamma is determined, the investor is assigned to a portfolio profile as follows:

$\boldsymbol{\gamma_{final}}$	**Profile**
$\gamma \geq 5.0$	Conservative
$5 > \gamma \geq 3.5$	Moderate
$\gamma < 3.5$	Aggressive

Gamma Assignment per Regime:

For each market regime, the model is intended to use different gamma values to adjust the level of risk according to the prevailing market conditions.

Regime	**Gamma**
Bull	$\gamma_{final} \times 0.30$
Transition	$\gamma_{final} \times 0.55$
Bear	$\gamma_{final} \times 1.00$

Profile Benchmark Allocations:

Portfolio allocation is also unique to each investor profile. The following benchmark allocations are used as the baseline for the regime detection model.

Profile	Equity	Bonds	Gold
Conservative	30%	60%	10%
Moderate	55%	40%	5%
Aggressive	85%	10%	5%

7.3 MODEL EXECUTION & CACHING

On the Run Model tab, the investor selects one of the 4 available models: Hidden Markov Model, Semi-Hidden Markov Model, Continuous Jump Model, or Machine Learning model. The selected model runs using the risk profile determined from the survey in Section 7.2.

Since some models are computationally expensive, we implemented a caching system to avoid running the same model more than once. Each time a model runs, a cache file is saved containing the model, risk profile, and results. If the investor returns and selects the same model with the same risk profile, they are prompted to load the existing cache file instead of rerunning the model, saving significant computation time.

If the investor chooses to run the model fresh, a progress bar and step-by-step log are displayed, showing the investor the current stage of the model execution in real time.

7.4 OVERVIEW TAB

The Overview tab provides a high-level summary of the model's performance:

- KPI cards at the top showing:
    - Total accumulated wealth based on the investor's initial investment amount
    - Annual return
    - Sharpe ratio
    - Sortino ratio
    - Maximum drawdown
- A cumulative return chart comparing the selected model against both the static benchmark and the MVO benchmark over the full testing period from 2019 to 2025

- Chart background shaded by detected regime — Bull, Transition, or Bear — allowing the investor to visually connect portfolio performance with the underlying market regime at each point in time

7.5 ANNUAL TAB

The Annual tab provides a year-by-year performance breakdown to validate the robustness of the model:

Charts comparing the model against both benchmarks on an annual basis:

- Annual return
- Annual Sharpe ratio
- Annual Sortino ratio
- Annual maximum drawdown

Monthly return heatmaps:

- Absolute monthly returns of the walk-forward model
- Monthly active return relative to the benchmark

The heatmaps allow the investor to identify which specific months and years the model added or lost value compared to the benchmark.

7.6 REGIMES TAB

The Regimes tab is the core analysis of the RAAM framework. The main goal of the model is to achieve better risk-adjusted returns through each regime on an aggregate basis. If the model fails to outperform the benchmark within individual regimes, it would indicate either that the regime detection is not effective or that the allocation is not efficient during each regime.

The tab displays 3 components:

- A regime attribution chart showing the return contribution by regime on an annual basis, allowing the investor to see the distribution of regimes annually
- A drawdown chart comparing the drawdown of the model against both benchmarks, showing the model's ability to protect capital during stress periods
- A regime comparison table summarizing for each detected regime:
    - Days in regime and % time
    - Model return and benchmark return
    - Alpha
    - Sharpe ratio and benchmark Sharpe
    - Delta Sharpe
    - Sortino ratio
    - Maximum drawdown

7.7 RISK TAB

The Risk tab includes a full risk metrics table covering all performance and risk measures to give the investor an overview of the model's performance and risk characteristics versus the benchmark. The tab also includes:

- VaR and CVaR charts at the 95% and 99% confidence levels
- Return distribution chart comparing the return profile of the model against the benchmark
- Full risk metrics table:

Metric	**Formula**	**Description**
Sharpe Ratio	$\frac{\mu \times 252}{\sigma \times \sqrt{252}}$	Return per unit of total risk
Sortino Ratio	$\frac{\mu \times 252}{\sigma_{down} \times \sqrt{252}}$	Return per unit of downside risk only
Calmar Ratio	$\frac{Annual\ Return}{Abs(Max\ DD)}$	Return per unit of maximum drawdown
Max Drawdown	$\min\left(\left(\frac{Cum_t - Peak_t}{Peak_t}\right)\right)$	Largest peak to trough decline
VaR (95%)	Percentile (r, 5%)	Worst expected daily loss at 95% confidence
CVaR (95%)	E [r given r <= VaR]	Average loss beyond the VaR threshold

7.8 TURNOVER TAB

The Turnover tab covers 2 charts:

- Annual turnover and number of switches per year
- Cumulative transaction cost over the full testing period

A detailed turnover table is also provided breaking down the switches, total turnover, average turnover per switch, and transaction cost by year.

7.9 WEIGHTS TAB

The Weights tab starts with a current allocation recommendation based on the model's detection of the current market regime, followed by a table showing the model allocation, benchmark allocation, and active tilts.

The tab also includes:

- A portfolio weights chart showing how the allocation changed over time from 2019 through 2025
- Average allocation per ETF across regimes, allowing the investor to understand how the model allocates differently during Bull, Transition, and Bear periods

7.10 COMPARE TAB

The Compare tab allows the investor to compare different cached model runs on the same chart. The user can select all available cached runs and compare them on the same cumulative wealth chart.

The tab also provides 2 summary tables:

- Performance summary table comparing the models across:
  - Annual return
  - Sharpe ratio
  - Sortino ratio
  - Maximum drawdown
  - Turnover
- Regime performance table comparing how each model performed during Bull, Transition, and Bear regimes specifically

7.11 EXPORT TAB

The Export tab allows the investor to generate a personalized client report in PDF format. The report includes 4 pages:

- Page 1: Thank you page with client summary including name, risk profile, model selected, and initial investment amount
- Page 2: Risk profile summary showing the investor's gamma parameters per regime and static benchmark allocation
- Page 3: Backtest performance overview including key metrics table and regime alpha attribution
- Page 4: Current allocation recommendation based on the latest regime detected by the model, showing model weights, benchmark weights, and active tilts per ETF

8. CONCLUSION

This paper demonstrates that compared to a static benchmark, a regime-aware allocation is able to produce better risk-adjusted returns. According to a 2019-2025 out of sample evaluation, the Machine Learning (ML) and Continuous Jump Model (CJM) both outperformed the static 55/30/5 on Sharpe and Sortino ratios, with ML Model showing a much better control on the drawdown and CJM achieving a lower turnover through fewer but more deliberate regime transitions.

The investor gets to decide the which models suits their need the best with risk-tolerant investors choosing CJM because of its low transaction costs stable allocations while risk-averse investors could choose ML models owing to their faster reaction to changes in regimes and their inherent lower exposures in downturn markets. Even though based in theory, the HMM and the HSMM were less competititve out of sample given the difficulty of tanslating probabilistic latent state models into consistent portfolio alpha generation.

One major limitation is the rate-shock regime of 2022 when the historically observed co-movement of bond and equities which was not being captured by models which were trained on historical data. These kinds of regimes are better detected when we have more observations from such novel states, causing models to better adapt and eventually improve regime detection.

BIBLIOGRAPHY

- Baitinger, Eduard, and Leonard Hoch. "Simplicity versus Complexity: A Comparative Analysis of HMM and HSMM for Regime-Based Asset Allocation." Available at SSRN 4796238 (2024).
- Qin, Shanshan, Zhenni Tan, and Yuehua Wu. "On robust estimation of hidden semi-Markov regime-switching models." Annals of Operations Research 338, no. 2 (2024): 1049-1081.
- Aydınhan, Afsar Onat, Petter N. Kolm, John M. Mulvey, and Yizhan Shu. 2024. "Identifying Patterns in Financial Markets: Extending the Statistical Jump Model for Regime Identification." Annals of Operations Research. https://doi.org/10.1007/s10479-024-06035-z.
- Baltussen, Guido, Laurens Swinkels, Bart van Vliet, and Pim van Vliet. 2023. "Investing in Deflation, Inflation, and Stagflation Regimes." Financial Analysts Journal 79 (3): 1-20.
- Nystrup, Peter, Henrik Madsen, and Erik Lindstrom. 2017. "Dynamic Portfolio Optimization Across Hidden Market Regimes." Quantitative Finance 18 (1): 83-95.
- Wood, Kieran, Stephen J. Roberts, and Stefan Zohren. 2026. "DeePM: Regime-Robust Deep Learning for Systematic Macro Portfolio Management." arXiv preprint arXiv:2601.05975.
- Rabiner, Lawrence R. “A Tutorial on Hidden Markov Models and Selected Applications in Speech Recognition.” Proceedings of the IEEE 77, no. 2 (1989): 257–286.
- Yu, Shiqi. “Hidden Semi-Markov Models.” Artificial Intelligence 174, no. 2 (2010): 215–243.

APPENDIX A: FULL FEATURE LIST

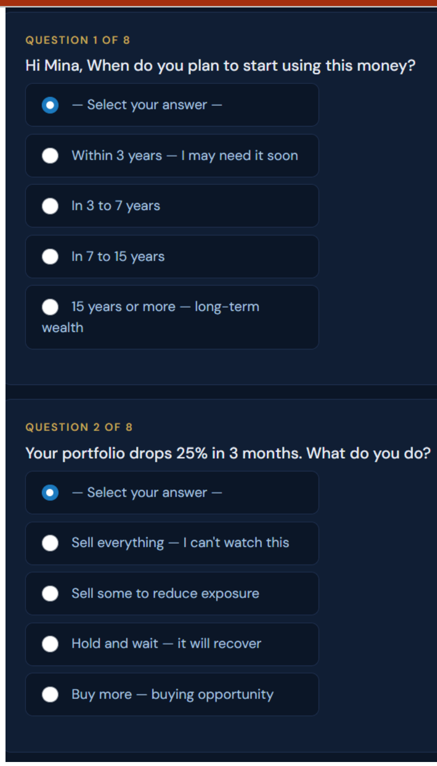
Bucket	Feature	Equation
Equity Market Dynamics	SPX Return	$log\left(\frac{SPX_t}{SPX_{t-1}}\right)$
Equity Market Dynamics	SPX Volatility (20d)	$std(SPX\ Return, 20d\ rolling)$
Equity Market Dynamics	SPX Drawdown	$\frac{SPX_t}{rolling_{max(SPX,252d)}} - 1$
Equity Market Dynamics	SPX MA Signal	$MA50 - MA200$
Equity Market Dynamics	Momentum Factor	$Fama - French\ Momentum$
Volatility / Risk Sentiment	VIX Level	VIX_t
Volatility / Risk Sentiment	VIX Change	$log\left(\frac{VIX_t}{VIX_{t-1}}\right)$
Volatility / Risk Sentiment	VIX Term Structure	$VIX \times \sqrt{3} - VIX$
Interest Rate Environment	10Y–2Y Slope	$Yield_{10Y} - Yield_{2Y}$
Interest Rate Environment	30Y–10Y Slope	$Yield_{30Y} - Yield_{10Y}$
Interest Rate Environment	Yield Change	$Yield_{10Y,t} - Yield_{10Y,t-1}$
Interest Rate Environment	Yield Curvature	$2 \times Yield_{10Y} - Yield_{2Y} - Yield_{30Y}$

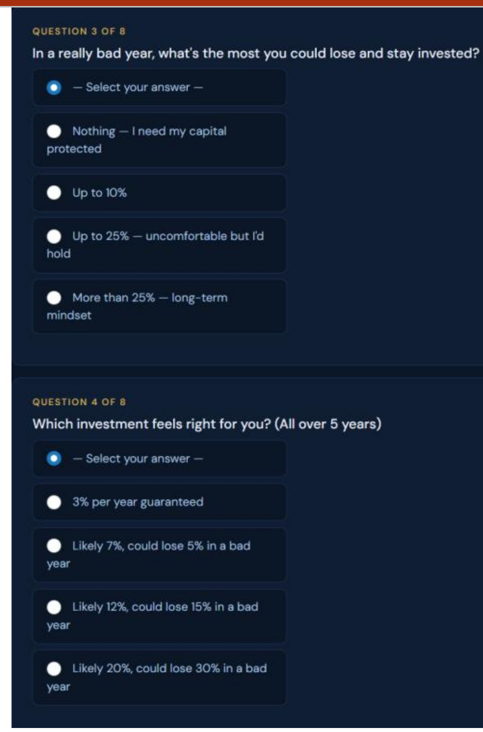
Cross-Asset Risk Signals	Equity–Bond Spread	$SPX\ Return\ -\ Bond\ Return$
Cross-Asset Risk Signals	Equity–Bond Correlation	$Corr(SPX\ Return, Bond\ Return, 60d\ rolling)$
Global Risk Environment Signals	DXY Return	$log\left(\frac{DXY_t}{DXY_{t-1}}\right)$
Global Risk Environment Signals	DXY Momentum (63d)	$\frac{DXY_t}{DXY_{t-63}} - 1$
Global Risk Environment Signals	DXY Volatility (21d)	$std(DXY\ Return, 21d\ rolling)$
Global Risk Environment Signals	DXY–SPX Correlation	$Corr(DXY\ Return, SPX\ Return, 63d\ rolling)$
Economic Activity Indicators	Industrial ISM (Lagged)	ISM_{t-1}
Economic Activity Indicators	ISM Change (Lagged)	$ISM_{t-1}\ -\ ISM_{t-2}$
Macro Inflation Environment	CPI Change (Lagged)	$CPI_{t-1}\ -\ CPI_{t-2}$
Macro Inflation Environment	CPI YoY (Lagged)	$\frac{CPI_{t-1}}{CPI_{t-13}} - 1$
Macro Inflation Environment	CPI Change 3M (Lagged)	$mean(CPI\ Change, 3m)$
Macro Inflation Environment	Inflation Regime	$1\ if\ CPI\ YoY\ >\ 3\%, else\ 0$
Macro Inflation Environment	UMich Sentiment (Lagged)	$UMich_{t-1}$

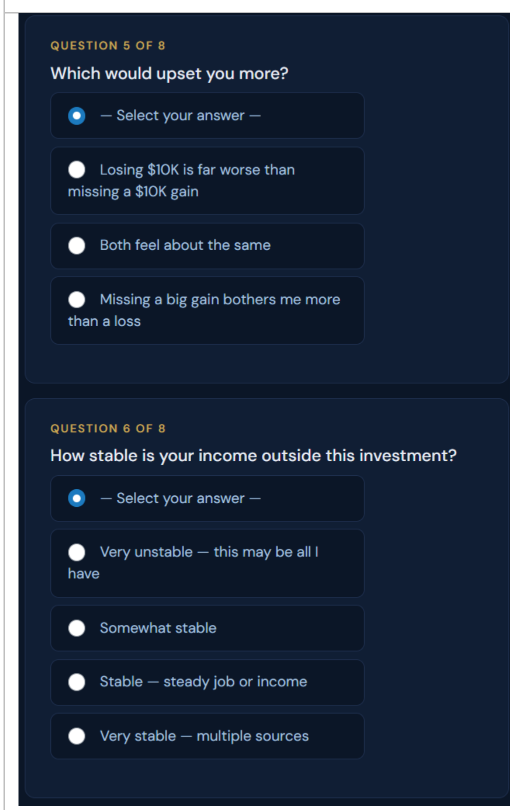
Macro Inflation Environment	Conference Board Confidence (Lagged)	CB_{t-1}
Liquidity / Monetary Conditions	M2 Growth (Lagged)	$\left(\frac{M2_t}{M2_{t-1}}\right) - 1$
Liquidity / Monetary Conditions	M2 Growth 3M (Lagged)	$\left(\frac{M2_t}{M2_{t-3}}\right) - 1$
Liquidity / Monetary Conditions	M2 YoY Growth (Lagged)	$\left(\frac{M2_t}{M2_{t-12}}\right) - 1$
Safe Haven Signals	Gold Return	$log\left(\frac{Gold_t}{Gold_{t-1}}\right)$
Safe Haven Signals	Gold/SPX Ratio	$\frac{Gold_t}{SPX_t}$
Safe Haven Signals	Gold/SPX Ratio Return	$log\left(\frac{\left(\frac{Gold}{SPX}\right)_t}{\left(\frac{Gold}{SPX}\right)_{(t-1)}}\right)$
Safe Haven Signals	Gold/SPX Momentum (21d)	$\frac{\left(\frac{Gold}{SPX}\right)_t}{\left(\frac{Gold}{SPX}\right)_{t-21}} - 1$
Safe Haven Signals	Gold/SPX Volatility (21d)	$std(\frac{Gold}{SPX}, 21d\ rolling)$
Commodity Inflation Shock	Oil Return	$log\left(\frac{Oil_t}{Oil_{t-1}}\right)$
Commodity Inflation Shock	Oil Volatility (21d)	$std(Oil\ Return, 21d\ rolling)$
Commodity Inflation Shock	Oil Momentum (63d)	$\frac{Oil_t}{Oil_{t-63}} - 1$
Bond Market Features	10Y Bond Return	$Bond\ Return_t$
Bond Market Features	10Y Bond Volatility	$std(Bond\ Return, 21d\ rolling)$

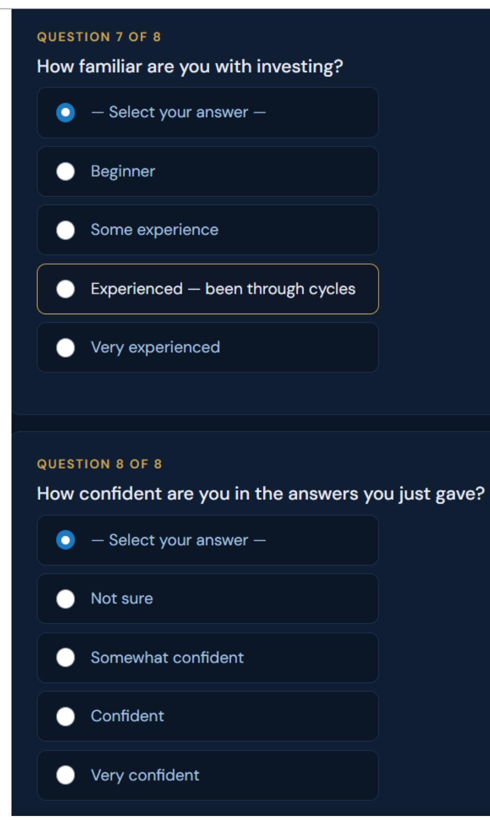
Credit Market Stress	IG Spread	IG_t
Credit Market Stress	HY Spread	HY_t
Credit Market Stress	HY Change	$HY_t - HY_{t-1}$
Credit Market Stress	IG Change	$IG_t - IG_{t-1}$
Credit Market Stress	HY–IG Spread	$HY_t - IG_t$
Recession Indicator	NBER Recession (Lagged)	$NBER_{t-1}$
Volatility / Risk Sentiment	VIX Raw	VIX_t
Cross-Asset Risk Signals	Bond Proxy Return	IEF_t
Global Risk Environment Signals	DXY Rolling Correlation	$Corr(DXY, SPX)$

APPENDIX B: RISK PROFILING SURVEY









APPENDIX C: WALK-FORWARD FOLD DETAILS

Fold	Training Period End	Prediction Period	Training Days	Prediction Days
1	2018-12-31	2019-01-01 to 2019-12-31	4,443	252
2	2019-12-31	2020-01-01 to 2020-12-31	4,695	253
3	2020-12-31	2021-01-01 to 2021-12-31	4,948	252
4	2021-12-31	2022-01-01 to 2022-12-31	5,200	251
5	2022-12-31	2023-01-01 to 2023-12-31	5,451	250
6	2023-12-31	2024-01-01 to 2024-12-31	5,701	252
7	2024-12-31	2025-01-01 to 2025-12-31	5,953	250

APPENDIX D: GMM REGIME STATISTICS

regime	0	1	2	3	4
equity_market_dynamics_features__SPX_return	-0.0015	-0.0011	0.0011	0.0007	0.0010
Volatility_Risk_Sentiment_features__VIX	31.5064	25.3835	17.5770	14.5928	16.1602
credit_features__HY_IG_Spread	5.2940	5.7435	3.7462	2.2665	2.7926
Macro_Inflation_Environment_features__CPI_change_3m_lag1	0.6264	0.4964	0.3911	0.6410	0.4582
Interest_Rate_Environment_features__Slope_10_2	0.0096	0.0165	0.0215	0.0005	0.0108
CrossAsset_RiskSignals_features__Equity_bond_corr	-0.2834	-0.4624	-0.4586	0.0149	-0.2961

APPENDIX E: FULL RISK METRICS TABLE

E.1 RISK PROFILE: MODERATE

Risk Metrics	Static BM	ML	Jump	HMM	HSMM
Ann. Return	8.38%	13.40%	13.78%	10.99%	11.08%
Ann. Vol	12.43%	12.35%	12.82%	13.15%	14.72%
Sharpe	0.71	1.081	1.072	0.859	0.788
Sortino	0.883	1.463	1.446	1.113	1.027
Calmar	0.374	0.541	0.464	0.385	0.394
Max DD	-22.40%	-24.80%	-29.70%	-28.50%	-28.10%
VaR (95%)	-1.12%	-1.25%	-1.26%	-1.34%	-1.39%
CVaR (95%)	-1.85%	-1.85%	-1.91%	-2.04%	-2.20%
Skewness	-0.418	-0.295	-0.112	-0.3	-0.266
Excess Kurtosis	12	2.946	5.097	3.922	7.343
Win Rate	66.70%	66.70%	70.20%	64.30%	65.50%
Switches	0	34	8	31	25
Sharpe Alpha	0	0.371	0.362	0.15	0.078

E.2 RISK PROFILE: CONSERVATIVE

Risk Metrics	Static BM	ML	Jump	HMM	HSMM
Ann. Return	5.61%	11.12%	11.59%	9.15%	7.46%
Ann. Vol	8.87%	10.13%	11.69%	10.82%	12.30%
Sharpe	0.659	1.092	0.997	0.864	0.647
Sortino	0.848	1.518	1.391	1.131	0.871
Calmar	0.255	0.499	0.4	0.326	0.258

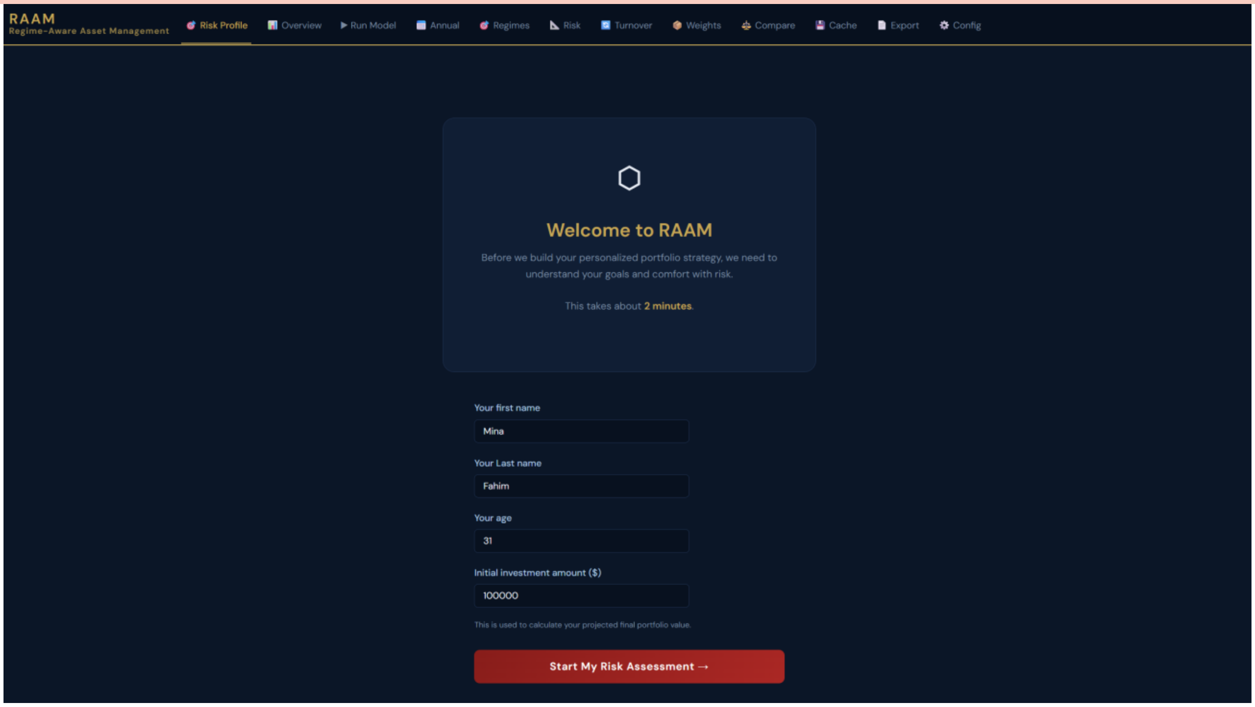
Max DD	-22.00%	-22.30%	-29.00%	-28.00%	-29.00%
VaR (95%)	-0.85%	-1.01%	-1.14%	-1.10%	-1.22%
CVaR (95%)	-1.31%	-1.48%	-1.70%	-1.65%	-1.81%
Skewness	-0.292	-0.226	0.074	-0.272	-0.089
Excess Kurtosis	8.964	2.546	4.961	3.799	7.057
Win Rate	63.10%	66.70%	67.90%	66.70%	59.50%
Switches	0	34	8	31	25
Sharpe Alpha	0	0.432	0.338	0.204	-0.013

E.3 RISK PROFILE: AGGRESSIVE

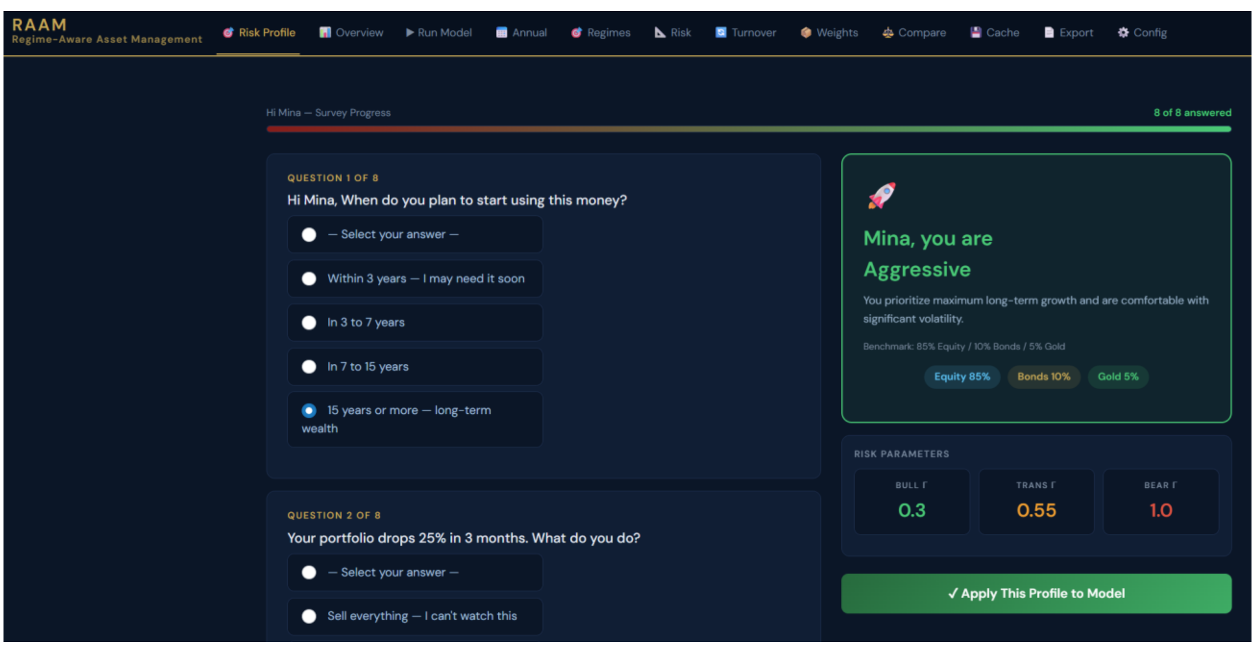
Risk Metrics	Static BM	ML	Jump	HMM	HSMM
Ann. Return	13.26%	13.98%	16.31%	12.44%	12.68%
Ann. Vol	17.43%	14.90%	14.59%	15.33%	16.42%
Sharpe	0.802	0.953	1.109	0.842	0.809
Sortino	0.988	1.248	1.423	1.055	1.035
Calmar	0.432	0.597	0.557	0.427	0.457
Max Drawdown	-30.70%	-23.40%	-29.30%	-29.10%	-27.70%
VaR (95%)	-1.56%	-1.51%	-1.39%	-1.45%	-1.55%
CVaR (95%)	-2.62%	-2.24%	-2.17%	-2.37%	-2.46%
Skewness	-0.4	-0.376	-0.287	-0.387	-0.352
Excess Kurtosis	13.363	5.514	8.263	6.668	8.577
Win Rate	66.70%	67.90%	71.40%	69.00%	66.70%
Switches	0	34	8	31	25
Sharpe Alpha	0	0.151	0.307	0.04	0.007

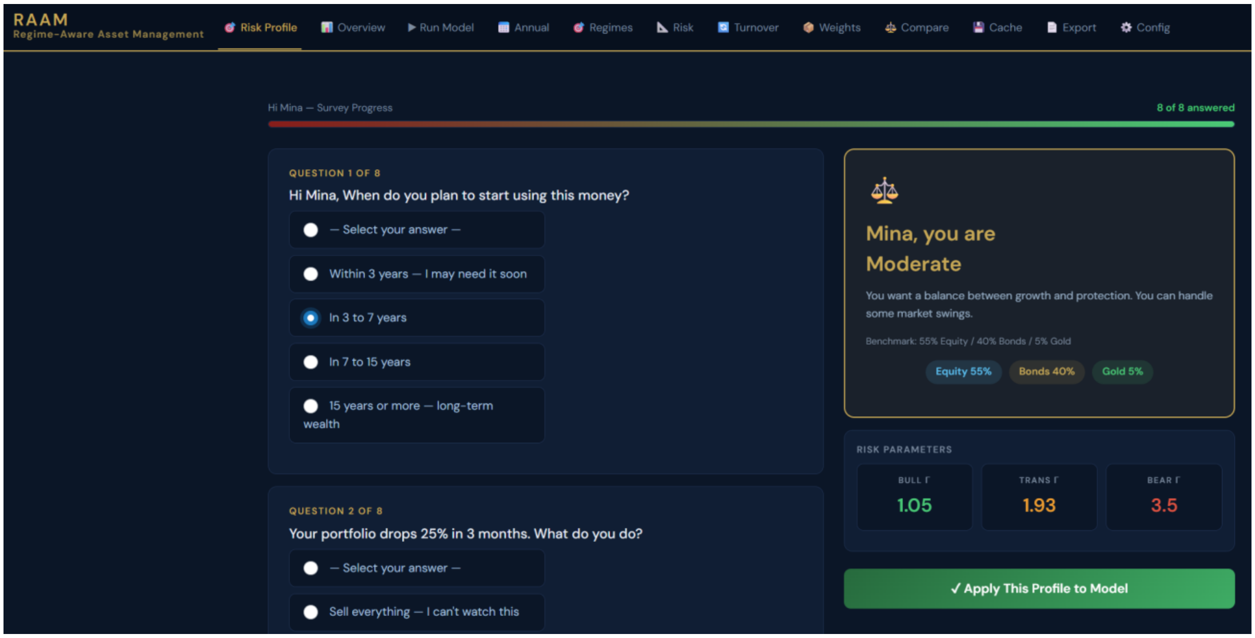
APPENDIX F: DASHBOARD SCREENSHOTS

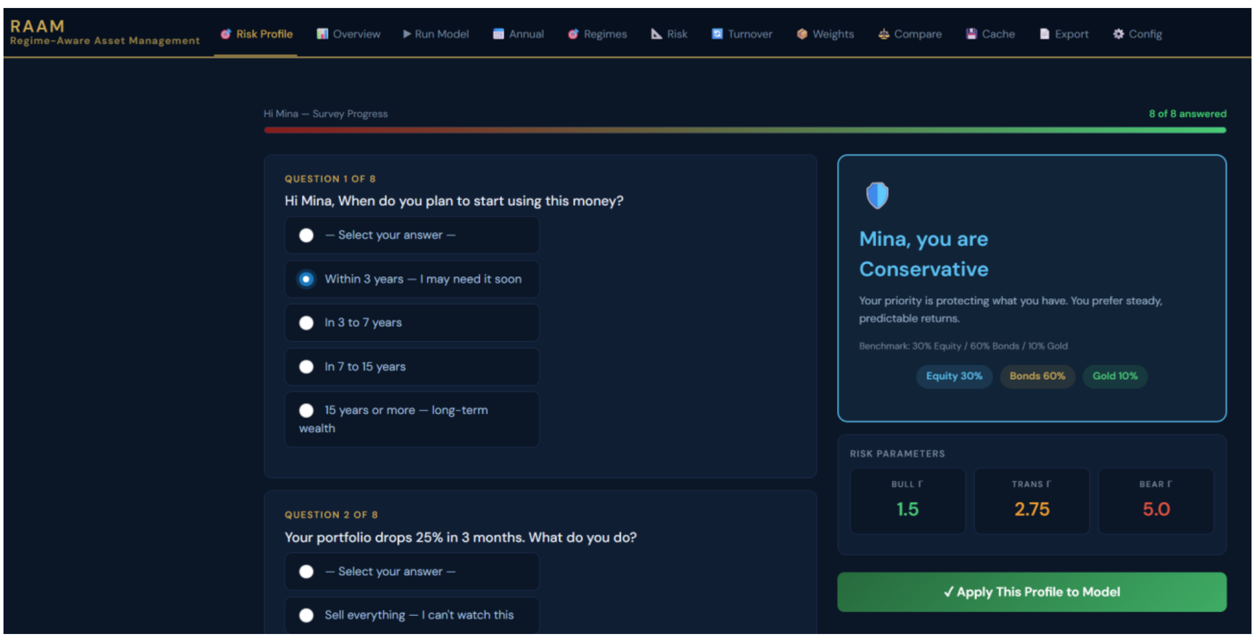
F.1 WELCOME PAGE



F.2 SURVEY AND RISK PROFILING




RAAM
Regime-Aware Asset Management
Risk Profile
Overview
Run Model
Annual
Regimes
Risk
Turnover
Weights
Compare
Cache
Export
Config
Hi Mina — Survey Progress
8 of 8 answered
QUESTION 1 OF 8
Hi Mina, When do you plan to start using this money?
— Select your answer —
Within 3 years — I may need it soon
In 3 to 7 years
In 7 to 15 years
15 years or more — long-term wealth
QUESTION 2 OF 8
Your portfolio drops 25% in 3 months. What do you do?
— Select your answer —
Sell everything — I can't watch this
Mina, you are Moderate
You want a balance between growth and protection. You can handle some market swings.
Benchmark: 55% Equity / 40% Bonds / 5% Gold
Equity 55%
Bonds 40%
Gold 5%
RISK PARAMETERS
BULL Γ
1.05
TRANS Γ
1.93
BEAR Γ
3.5
✓ Apply This Profile to Model


RAAM
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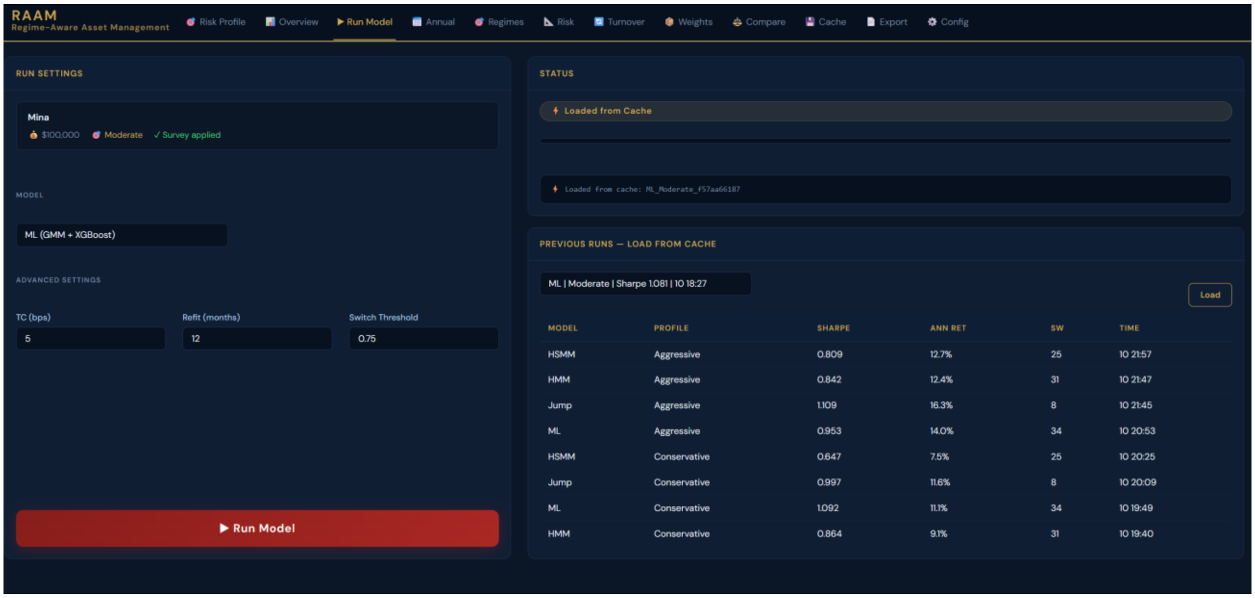
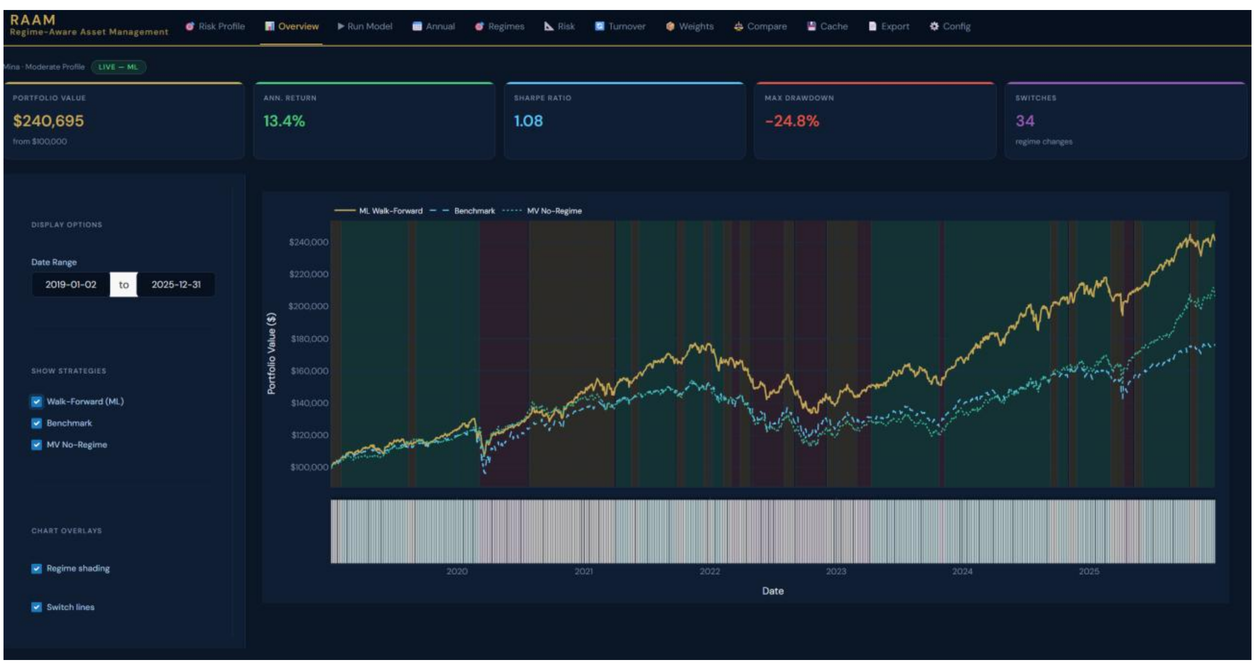
Hi Mina — Survey Progress
8 of 8 answered
QUESTION 1 OF 8
Hi Mina, When do you plan to start using this money?
— Select your answer —
Within 3 years — I may need it soon
In 3 to 7 years
In 7 to 15 years
15 years or more — long-term wealth
QUESTION 2 OF 8
Your portfolio drops 25% in 3 months. What do you do?
— Select your answer —
Sell everything — I can't watch this
Mina, you are Conservative
Your priority is protecting what you have. You prefer steady, predictable returns.
Benchmark: 30% Equity / 60% Bonds / 10% Gold
Equity 30%
Bonds 60%
Gold 10%
RISK PARAMETERS
BULL Γ
1.5
TRANS Γ
2.75
BEAR Γ
5.0
✓ Apply This Profile to Model

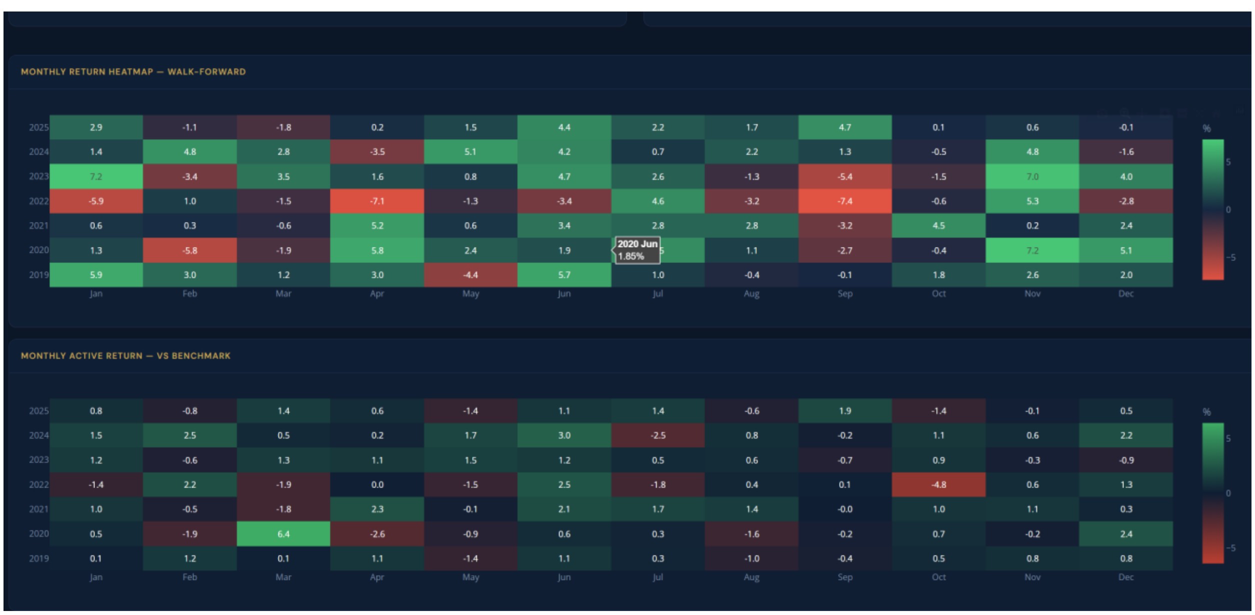
F.3 CHOOSE AND RUN MODEL



F.4 MODEL OVERVIEW



F.5 ANNUAL PERFORMANCE ANALYSIS

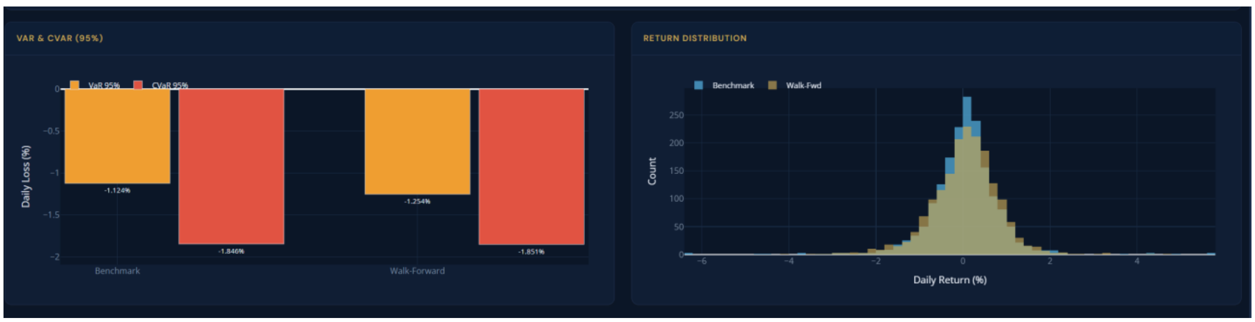


F.6 REGIME ATTRIBUTION ANALYSIS

REGIME	DAYS	% TIME	MODEL RET	BMARK RET	ALPHA	SHARPE	BM SHARPE	MAX DD	BM MAX DD
Bull	988	56.1%	15.96%	8.20%	+7.77%	1.349	1.000	-10.0%	-9.5%
Transition	467	26.5%	8.56%	4.48%	+4.08%	0.710	0.420	-17.6%	-19.4%
Bear	305	17.3%	12.78%	15.24%	-2.47%	0.907	0.777	-11.8%	-18.1%

F.7 RISK ANALYSIS

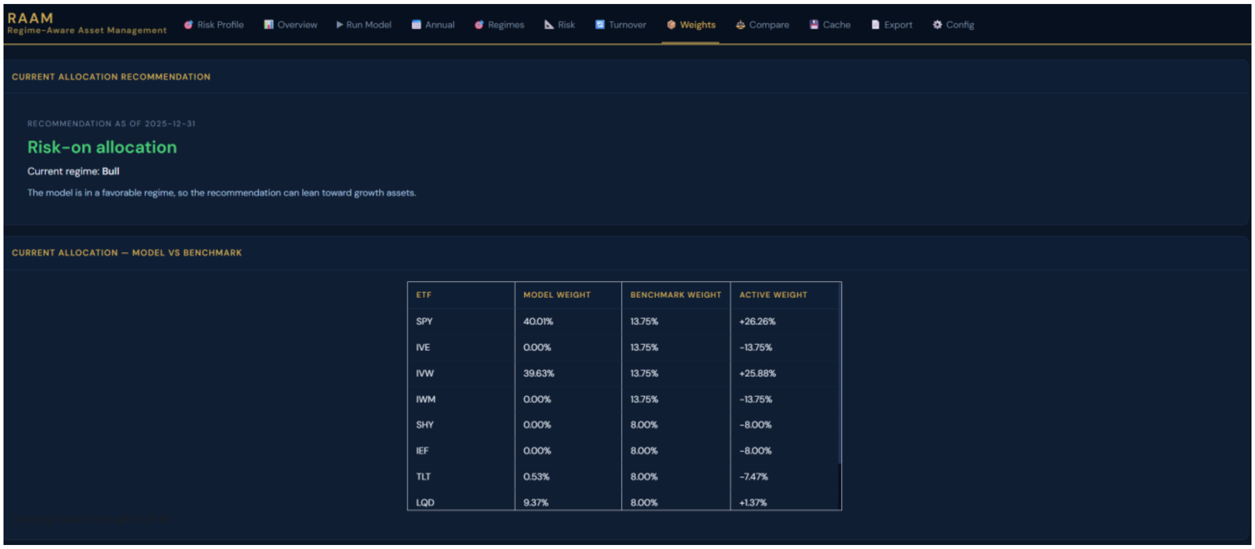
METRIC	BENCHMARK	WALK-FORWARD
Ann Vol	12.43%	12.35%
Sharpe	0.710	1.081
Sortino	0.664	1.011
Calmar	0.374	0.541
Max DD	-22.4%	-24.8%
VaR 95%	-1.124%	-1.254%
CVaR 95%	-1.846%	-1.851%
VaR 99%	-2.002%	-2.146%
CVaR 99%	-3.367%	-2.749%
Skewness	-0.418	-0.295
Ex Kurtosis	11.962	2.934
Omega	1.143	1.202
Pain Index	5.340%	5.234%
Win Rate	66.7%	66.7%



F.8 TURNOVER ANALYSIS

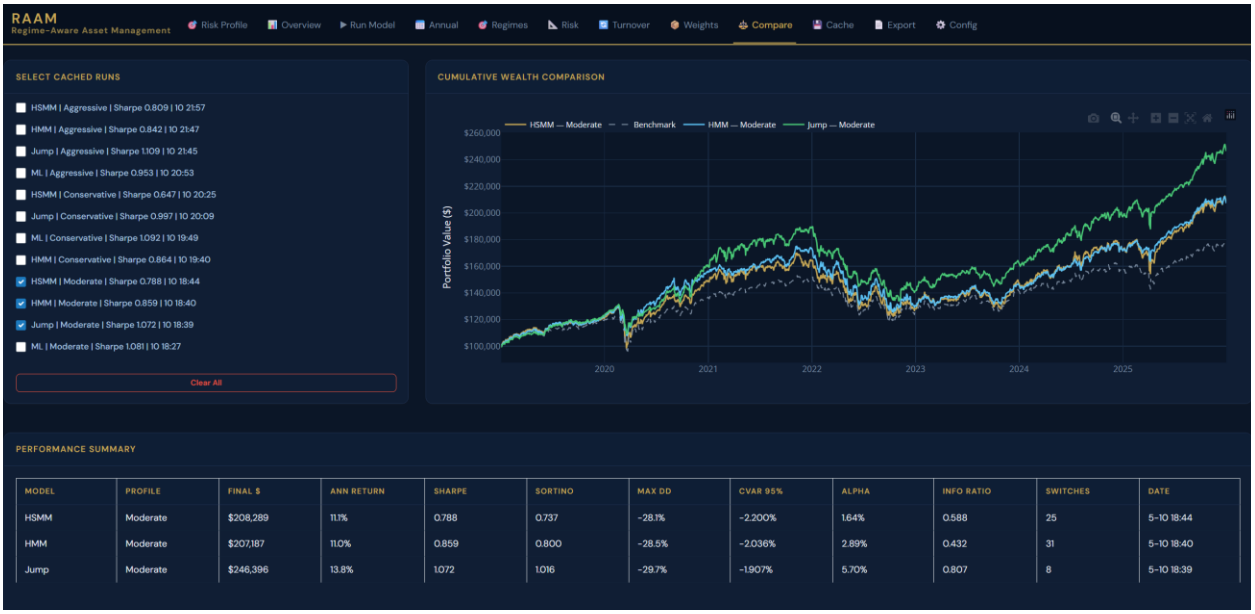
YEAR	SWITCHES	TURNOVER	AVG/SWITCH	TC (BPS)
2019	4	4.898	1.225	24.5
2020	2	2.326	1.163	11.6
2021	6	8.288	1.381	41.4
2022	8	7.884	0.986	39.4
2023	4	4.542	1.135	22.7
2024	4	6.727	1.682	33.6
2025	6	8.031	1.338	40.2

F.9 RECOMMENDED ALLOCATION

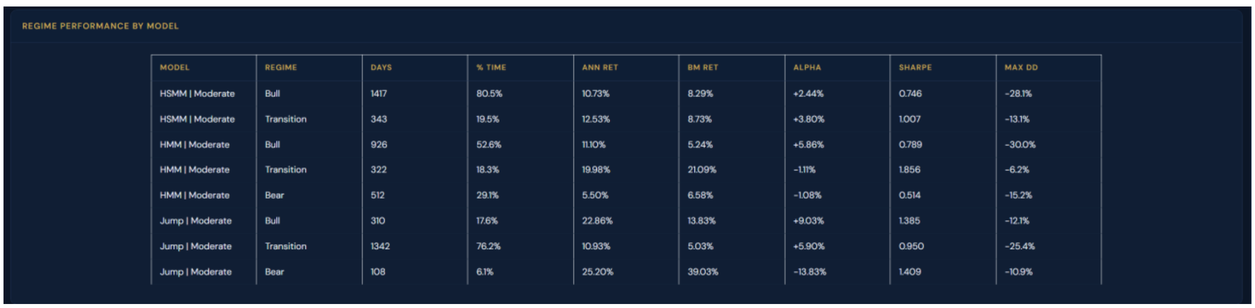


ETF	MODEL WEIGHT	BENCHMARK WEIGHT	ACTIVE WEIGHT
SPY	40.01%	13.75%	+26.26%
IVE	0.00%	13.75%	-13.75%
IVW	39.63%	13.75%	+25.88%
IWM	0.00%	13.75%	-13.75%
SHY	0.00%	8.00%	-8.00%
IEF	0.00%	8.00%	-8.00%
TLT	0.53%	8.00%	-7.47%
LQD	9.37%	8.00%	+1.37%

F.10 MODEL COMPARISON



MODEL	PROFILE	FINAL $	ANN RETURN	SHARPE	SORTINO	MAX DD	CVAR 95%	ALPHA	INFO RATIO	SWITCHES	DATE
HSMM	Moderate	$208,289	11.1%	0.788	0.737	-28.1%	-2.200%	1.64%	0.588	25	5-10 18:44
HMM	Moderate	$207,187	11.0%	0.859	0.800	-28.5%	-2.036%	2.89%	0.432	31	5-10 18:40
Jump	Moderate	$246,396	13.8%	1.072	1.016	-29.7%	-1.907%	5.70%	0.807	8	5-10 18:39

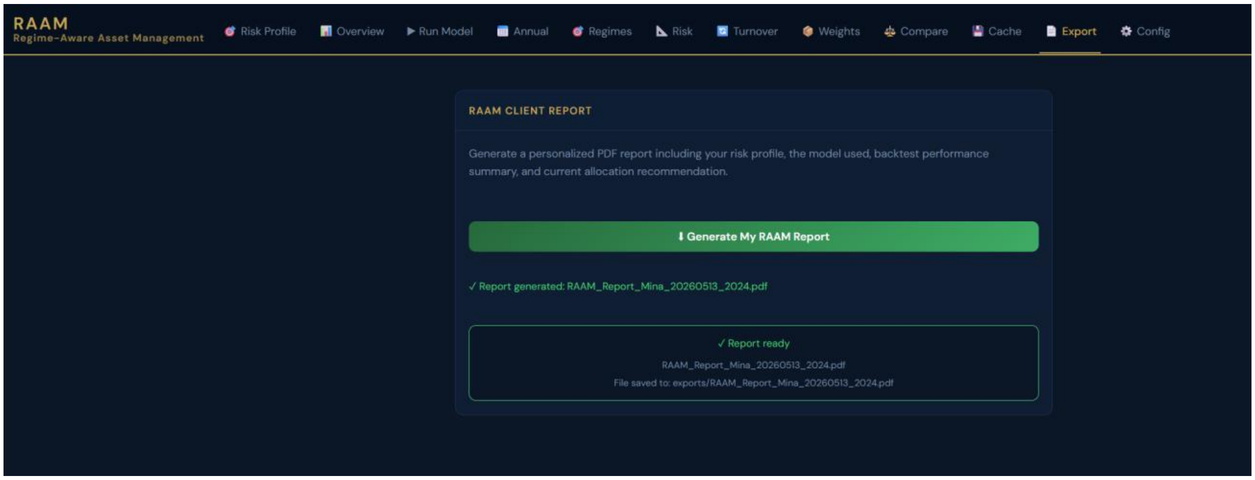


MODEL	REGIME	DAYS	% TIME	ANN RET	BM RET	ALPHA	SHARPE	MAX DD
HSMM | Moderate	Bull	1417	80.5%	10.73%	8.29%	+2.44%	0.746	-28.1%
HSMM | Moderate	Transition	343	19.5%	12.53%	8.73%	+3.80%	1.007	-13.1%
HMM | Moderate	Bull	926	52.6%	11.10%	5.24%	+5.86%	0.789	-30.0%
HMM | Moderate	Transition	322	18.3%	19.98%	21.09%	-1.11%	1.856	-6.2%
HMM | Moderate	Bear	512	29.1%	5.50%	6.58%	-1.08%	0.514	-15.2%
Jump | Moderate	Bull	310	17.6%	22.86%	13.83%	+9.03%	1.385	-12.1%
Jump | Moderate	Transition	1342	76.2%	10.93%	5.03%	+5.90%	0.950	-25.4%
Jump | Moderate	Bear	108	6.1%	25.20%	39.03%	-13.83%	1.409	-10.9%

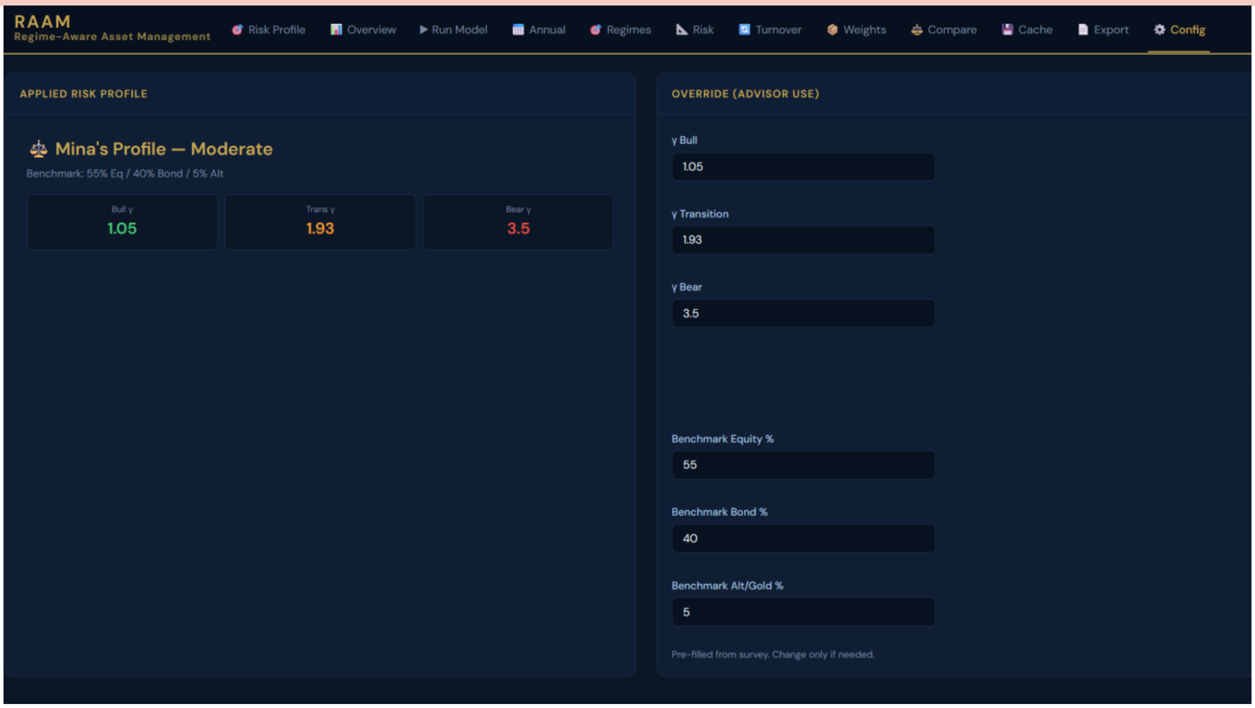
F.11 CACHE MANAGEMENT

MODEL	PROFILE	SHARPE	ANN RET	SW	RUNTIME(S)	DATE	MB	KEY
HSMM	Aggressive	0.809	12.7%	25	453.1	2026-05-10 21:57	0.3	HSMM_Aggressive_1b376820dc
HMM	Aggressive	0.842	12.4%	31	57.4	2026-05-10 21:47	0.3	HMM_Aggressive_8cef21daf0
Jump	Aggressive	1.109	16.3%	8	773.4	2026-05-10 21:45	0.3	Jump_Aggressive_c4edbb0158
ML	Aggressive	0.953	14.0%	34	209.1	2026-05-10 20:53	0.3	ML_Aggressive_e74f88ad7e
HSMM	Conservative	0.647	7.5%	25	431.6	2026-05-10 20:25	0.3	HSMM_Conservative_855bed7ed1
Jump	Conservative	0.997	11.6%	8	513.5	2026-05-10	0.3	Jump_Conservative_32d4a9e7fa

F.12 CLIENT REPORT GENERATION



F.13 CONFIGURATION



APPENDIX G: CLIENT REPORT SAMPLE

RAAM
Regime-Aware Asset Management

Thank you for using the RAAM Platform, Mina.

Your personalized portfolio report has been generated based on your risk profile and the regime-aware model you selected. This report covers your risk profile summary, backtest performance overview, and current allocation recommendation.

Report Date	**May 13, 2026**
Client Name	**Mina**
Risk Profile	**Moderate**
Model Selected	**ML**
Initial Investment	**$100,000**

1. Your Risk Profile

Moderate

You seek a balance between growth and protection with manageable volatility. The model balances equity and bond exposure dynamically across regimes.

Risk Aversion Parameters

Regime	Bull	Transition	Bear
Risk Aversion (γ)	1.05	1.93	3.5

Static Benchmark Allocation

Asset Class	Benchmark Weight
Equity	55%
Bonds	40%
Gold	5%

2. Backtest Performance Overview

Model: ML Period: 2019 - 2025 Method: Walk-Forward Validation

Key Performance Metrics

Metric	RAAM Model	Benchmark
Final Value	$240,695	$175,379
Ann. Return	13.40%	8.38%
Sharpe Ratio	1.081	0.710
Sortino Ratio	1.011	0.664
Calmar Ratio	0.541	0.374
Max Drawdown	-24.8%	-22.4%
Sharpe Alpha	0.371	0.000
Regime Switches	34	0

Regime Alpha Attribution

Regime	Days	% Time	Model Ret	BM Ret	Alpha	Sharpe
Bull	988	56.1%	15.96%	8.20%	+7.77%	1.349
Transition	467	26.5%	8.56%	4.48%	+4.08%	0.710
Bear	305	17.3%	12.78%	15.24%	-2.47%	0.907

3. Recommended Allocation

Based on the latest regime detected by the model, the following portfolio allocation is recommended:

Current Detected Regime: Bull

ETF	RAAM Weight	Benchmark	Active Tilt
SPY	40.0%	13.8%	+26.3%
IVW	39.6%	13.8%	+25.9%
GLD	10.5%	5.0%	+5.5%
LQD	9.4%	8.0%	+1.4%
TLT	0.5%	8.0%	-7.5%
IWM	0.0%	13.8%	-13.8%
IVE	0.0%	13.8%	-13.8%
SHY	0.0%	8.0%	-8.0%
IEF	0.0%	8.0%	-8.0%
HYG	0.0%	8.0%	-8.0%

RAAM Platform | Stevens Institute of Technology — FE800 | May 13, 2026 | For informational and educational purposes only. Not financial advice.